Measuring the Unified Intelligence Level:
A Cumulative, Harness-Measurable Hierarchy

Five Conceptual Intelligence Families, Six Measurable Coordinates, and a Gated U0–UΩ Intelligence-Level Scale

Version 2.0 — memory measured on its own axis,
and same-seed executor comparison outside the mechanically-derivable envelope

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Abstract

Version 2.0 result. This version adds a same-seed comparison of the archived frontier and Haiku executors on sealed regime-switch discovery. With harness and protocol fixed, frontier passes 5/12 (41.7%) and Haiku 1/12 (8.3%); the exact two-sided McNemar test is $p = 0.125$, so the binary difference is not claimed as significant. Frontier has lower RMSE on 11/12 matched instances (exact two-sided sign test $p = 0.00635$), with medians 0.639 versus 5.112 and nearest-neighbour-normalized medians 0.359 versus 2.142. All 24 episodes exit 0 and state a mechanism. This supports discrimination on these instances, not universal model-tier ordering, and is not a Harness Scaling Curve segment.

Artificial intelligence is often summarized with a single capability label, yet several distinct properties are routinely conflated: raw cognitive problem solving, persistent individual learning and self-improvement, collective organizational intelligence, autonomous task completion, and evidence-grounded self-modeling. This paper proposes a unified framework that separates these properties while preserving a readable ordinal hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is not treated as an irreducible scalar; it is measured by two primary coordinates, Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p. The theoretical framework therefore contains five conceptual families but six directly measured coordinates: $[C, I, O, T, H, SA]$. Computer-screen understanding is treated as a diagnostic subscale of Cognitive Intelligence rather than a sixth family, so that computer-agent failures remain legible without double-counting. Long-term Memory Capability M is measured independently as a supporting coordinate rather than identified with Individual Intelligence: memory remains a temporal substrate and a prerequisite for higher I-levels, but a high memory level is not sufficient for a high I-level and may exceed it. Formally $I(A) \geq I_n \implies M(A) \geq \mu_n$, while the converse does not hold, and M is excluded from the default HLIS product to avoid double counting. Individual Intelligence continues to require long-term memory as its temporal substrate rather than adding memory as a sixth family: I1 requires persistent episodic and task continuity across restarts, I2 adds semantic consolidation and procedural learning, I3–I4 require durable self-improvement and meta-improvement lineage, and I5 and above require hypothesis–experiment–evidence discovery memory. A long context window is therefore not long-term memory. A cumulative Unified Intelligence scale U0–UΩ is defined by gated promotion: a system receives level U_n only if it retains all lower-level capabilities and satisfies minimum requirements in C, I, O, SA and a required point on the delegation success surface $S_A(T, H)$. The framework explicitly rejects simple averaging, because strong performance in one dimension must not conceal a

critical weakness in another. It also separates intelligence from substrate reach, authority, write depth, verification, resources, and physical constraints. To make the framework useful for model and harness comparison, it defines a Harness-LLM Intelligence Score (HLIS) for one concrete LLM-harness pair and a Harness-Intelligence-Level (HIL) characterization for a base LLM evaluated across a standardized ladder of harness generations.

New in Version 2.0. Version 2.0 carries forward Version 1.9's item-validity and apparatus findings and adds a same-seed regime-switch comparison: the common protocol is run with the archived frontier and Haiku executors, changing the executor/model while holding the harness, task family and seeds fixed (§16.10).

The measurement work continues the search for a difficulty axis that reaches the frontier, and adds two findings about the instrument rather than any model. First, **a fourth answer-key defect**, in the axis built specifically to be different: the decidability class appeared to catch the frontier and was catching two rules of the author's that read as precedence statements. Four defects in four families, each found only when a result looked surprising, is now this paper's strongest evidence for auditing items and its strongest caution about how weak the audit's detection rate is. Second, **a timeout that could not fire**: the executor harness killed only its direct child and then waited on a pipe held by grandchildren, so a run exceeding its limit hung silently and the episode that eventually returned was scored as an empty delivery. A measurement apparatus that cannot distinguish a killed run from a failed one produces data that looks clean and is not.

Version 1.9 reported the first live results from outside the mechanically-derivable envelope. Version 2.0 extends that result with a matched executor/model comparison: frontier passes 5/12 while Haiku passes 1/12 on the same twelve seeds (§16.10). The comparison is deliberately not a Harness Scaling Curve segment, because HSC holds a frozen model fixed and varies harness generation.

Carried forward from Version 1.8. Version 1.7 added computer-screen understanding as a diagnostic subscale of Cognitive Intelligence, and that is carried forward here (§4.1): GUI Perception is placed inside C rather than made a sixth family, task difficulty gains a perceptual component, and closed-loop computer use is evaluated inside DI.

The measurement work carried forward in this version is a search for a difficulty axis that reaches the frontier. Four axes have now been built and run: disclosed reasoning hazards, specification scale, exact search, and *decidability*. The first separates no model; the second separates executor tiers and not the frontier; the third and fourth separate nothing, with every executor at 1.000. The negative results support the mechanically-derivable envelope argument (§16.9). Version 2.0's contribution is narrower: it tests whether the sealed regime-switch family discriminates two executors under a common same-seed protocol, and reports the paired result with uncertainty rather than treating it as a harness curve.

Carried forward from Version 1.6. Versions 1.4 and 1.5 established what to measure. This version is about whether the instrument that measures it is sound. Building harder delegation items to relieve the saturation reported in Version 1.4 produced two findings. First, difficulty rather than trickiness is the lever: three items built around specific reasoning hazards separated a scripted-correct solver from a naive one and separated no *model*, because an item that signposts its own hazard is answered by anyone who reads it. Second, and more consequentially for anyone running a hidden-key benchmark, **two executors of very different capability returning the same answer to the same item is evidence that the item is self-contradictory, not that it is hard**. Two real defects were found this way in items that had passed their own test suite, and both had been scored as model failures. Version 1.6 therefore adds cross-tier concordance auditing as a required validity check on any hidden answer key (§16.3), requires that parameterised item families test their prose against their own key (§16.4), replaces binary item scoring with graded scoring and separated failure modes at the item level (§16.5), and reports the resulting measurements (§16.2). It also embeds long-term memory inside Individual Intelligence (§5.1) and carries forward the acceptance-invariance repair of Version 1.4.

Carried forward from Version 1.4. The ladder has been built and a curve has been measured. Four harness generations were instantiated and one frozen model was run across them under identical tasks, seeds, verifiers and intervention budget. The measurement produced a finding about the score rather than about the model: **the delegation achievement variable A_{DI} is exactly invariant to the harness**

generation that adds an acceptance step, because A_{DI} is defined on $P(\text{success})$ and acceptance does not change the probability of success — it changes what is reported as success. Measured, the first two rungs scored 93.3 and 93.3 while work handed back as finished and wrong went from 2/12 to 0/12. Version 1.4 therefore redefines the delegation primitive on *delivered outcomes* net of false completions (§13.9), requires the acceptance outcome to be recorded beside the verifier outcome (§13.12), adds verification responsiveness ρ_V as the term attributable to the model rather than the harness (§14.3), records two structural cautions about the composite HIL-Score (§14.4), and reports the first curve with its limitations (§17.4). Everything else is carried forward from Version 1.3 substantially unchanged.

Keywords: unified intelligence, AGI levels, cognitive intelligence, individual intelligence, organizational intelligence, delegation intelligence, self-awareness, task difficulty, human intervention, harness architecture, recursive self-improvement, harness intelligence level, harness-LLM score, harnessability, benchmark ladder, harness scaling curve, long-term memory, episodic memory, semantic consolidation, procedural memory, benchmark validity, answer-key defects, cross-tier concordance, item saturation, graded scoring, memory capability, memory prerequisite gate, GUI perception, computer-use agents, spatial grounding, benchmark headroom, escalation, underdetermined tasks, sealed-mechanism discovery

Contents

1 Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically: solving a difficult problem is not the same as learning from experience; learning is not the same as improving the learning mechanism; coordinating several agents is not the same as being a stronger individual; and completing a task independently is not the same as merely producing a strong answer when a human structures every step. Second, the taxonomy must remain readable enough that a system can be summarized with a level that people can understand.

This paper proposes a compromise between a single scalar and an unconstrained profile of dozens of metrics. The theory uses five conceptual intelligence families — C, I, O, DI and SA — but treats Delegation Intelligence as a measurable two-coordinate phenomenon rather than a primitive scalar. The resulting scientific profile uses six coordinates: C, I, O, T, H and SA, with verified reliability p attached to task-completion claims.

The central design principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus U4 is not simply a system that exhibits one advanced behavior associated with U4; it is a system that still passes U0–U3 retention tests and also passes the U4 gate. In this sense, higher U-levels denote broader validated intelligence envelopes, although they need not be faster, cheaper, or more accurate than lower-level specialized systems on every narrow task.

1.1 What changed in this version, and why

Versions 1.0 through 1.3 developed the framework analytically. Version 1.3 in particular made the scoring apparatus formal: it defined every symbol in HLIS, said how each achievement variable is constructed, introduced the cumulative $q^* = \min$ rule that prevents level skipping, and required that a dimension with no evidence be omitted rather than zeroed.

Version 1.4 differed in kind rather than in degree of detail. The harness ladder it specifies had been *built* to HG3 and a model run across it. A scoring formula can be argued about indefinitely and can only be found wrong by being run, and that one was.

Version 1.5 consolidated two refinements. It embedded long-term memory inside Individual Intelligence rather than adding Memory as a sixth family, on the grounds that memory capacity alone is not intelligence and a separate top-level score would double-count the persistence I already requires. And it generalised the acceptance finding into a standing requirement that delivered-correct reliability be reported separately from outcome integrity. Both are carried into this version, the first in §5.1.

Version 1.6 turns the same scepticism on the measuring instrument. Version 1.4 reported that the task suite was saturated — the stronger executors had no headroom, and rungs that differed materially produced identical curves. A curve through a saturated suite measures the suite. The obvious response is harder items, and building them is where this version's findings came from.

The Version 1.6 item-validity finding

An item with a hidden answer key can be wrong, and when it is, it does not look wrong. It looks like a hard item that models fail.

Two items in the delegation suite were self-contradictory: each stated one rule in its visible specification and graded a different one. Both had passed a test suite written for them. Both were discovered the same way — a frontier executor and a much weaker one returned the *same* answer to the *same* item, and both were marked wrong.

That signature is diagnostic. Independent systems of different capability do not usually agree on the answer to a question that is hard; they agree on the answer to a question that is clear, and if the key disagrees with both of them, the prior belongs on the key. §16.3 makes the check a requirement.

The general point is that every claim in this framework rests on items whose keys are asserted rather than proved. The harness ladder has a controlled comparison at its centre — the paired design of §17.4 — but no such control protects the items themselves. Version 1.6 supplies one.

The Version 1.4 scoring finding

A_{DI} is the weighted mean of $S_A(T,H) = P(\text{success})$. A harness generation whose entire contribution is an acceptance step does not change the probability of success — it changes which results are *reported* as successes. Such a generation therefore scores identically to the generation below it.

Measured on 48 episodes with one frozen model and only the harness changing, HG0 and HG1 both scored 93.3 while false completions went from 2/12 to 0/12.

The rung the curve could not see is the one on which the reportability of every rung above it depends, since a result accepted by its own producer is an assertion rather than a completed task. §13.9 repairs the primitive.

1.2 Contributions

1. A separation of five conceptual intelligence families into six directly measured coordinates, $[C, I, O, T, H, SA]$, with Delegation Intelligence decomposed rather than treated as a primitive scalar (§3–§7).
2. A cumulative, gated Unified Intelligence Level scale U0–UΩ, in which promotion requires retention of all lower levels and satisfaction of every coordinate gate (§8–§10).
3. A harness realization of the hierarchy, and the position that the (model, harness) pair is the basic experimental unit (§11).
4. A continuous pair score HLIS with a fully specified construction, and a harness-indexed characterization HIL of a base model across a standardized ladder (§13–§14).
5. The first instantiation of the ladder, the first measured harness scaling curves, and a proposition and repair concerning the delegation achievement variable that the measurement exposed (§11.3, §13.9, §17.4).
6. Long-term memory embedded inside Individual Intelligence as its temporal substrate, with a per-level memory contract and an explicit separation of long context from persistence (§5.1).
7. A validity protocol for benchmark items: cross-tier concordance auditing, specification–key consistency tests for parameterised families, and graded scoring with separated failure modes (§16).
8. **New in Version 2.0:** a same-seed executor/model comparison on the sealed regime-switch family, including paired pass outcomes, RMSE and nearest-neighbour-baseline analysis, with exact McNemar and sign-test results and explicit resource and variance limitations (§16.10).

1.3 Scope and status

This is a framework proposal with two distinct preliminary measurement programs, included because they changed or tested the instrument rather than because they settle any question about a system. The harness-ladder study in §17.4 spans three executors, 48 episodes per executor and one instrumented coordinate. The sealed-discovery study in §16.10 adds 24 matched executor–seed episodes on one mechanism family. The proposed levels and numerical thresholds still require calibration across domains, models and harnesses; §21 states the limitations in full.

2 Design Requirements for a Unified Scale

- **Orthogonality:** each core dimension should answer a materially different question.
- **Operationality:** every level should be testable through behavior, state, or externally verifiable outcomes rather than self-description.
- **Cumulative inheritance:** U_{n+1} must retain U_n capabilities; level skipping is not sufficient for certification.
- **Bottleneck sensitivity:** exceptional strength in one dimension must not hide a critical weakness in another.
- **Role neutrality:** authority, resources, substrate access, and safety policy must not be confused with intelligence itself.
- **Harness realizability:** each transition should correspond to an implementable architectural mechanism around an LLM or multi-agent system.
- **Harness comparability:** base LLMs should be evaluated on a standardized harness ladder so model quality and harness quality can be separated.
- **Dataset coupling:** every claimed level must have a matched hidden or externally verified benchmark dataset; architectural mechanisms alone do not earn a level.
- **Longitudinal validity:** learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time rather than one-shot demonstrations.
- **Temporal-substrate discipline (new in 1.5):** long-term memory is embedded inside Individual Intelligence rather than added as a separate top-level family. An I1 or higher claim requires evidence of persistence across episodes and process restarts, not retrieval within a single long context.
- **Item validity (new in 1.6):** a benchmark item with a hidden key is an unverified assertion until something checks it. Every item family must carry a test that its visible specification and its answer key agree, and any item that two executors of materially different capability fail identically must be audited before its failures are attributed to the executors.
- **Outcome completeness (new in 1.4):** a scoring variable must distinguish outcomes that differ in what they cost the person delegating. A refusal and a confidently wrong delivery are not the same failure, and a score that rates them equally will, given a gradient, select against the refusal.

3 Five Conceptual Families and Six Measurable Coordinates

$$\text{Conceptual core} = [C, I, O, DI, SA]$$
$$\text{Measured core} = [C, I, O, T, H, SA], \text{ with DI derived from } (T, H, p)$$

4 Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, autonomy, organization and self-modification. This distinction matters because a highly capable base model may reason at an expert level while remaining a transient tool, whereas a weaker model embedded in an excellent harness may complete long projects through memory, planning, retries and verification.

Family	Question answered	Primary object of measurement
C — Cognitive	How well can the system understand, reason, generalize, model and solve?	Problem-solving capability under controlled information and tool access.
I — Individual	How deeply can one persistent intelligence remember, learn, self-improve, recursively improve and discover?	Evolution of one persistent decision locus.
O — Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve and discover collectively?	Coordination structure plus member roles and evidence flow.
DI — Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Success surface over T and H at reliability p.
SA — Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes and role?	Evidence-grounded self-model and reflexive reasoning.

Level	Name	Operational interpretation
C0	Reactive	Direct pattern-conditioned response on simple problems.
C1	Contextual	Understands ordinary context; solves routine, well-specified problems.
C2	Compositional	Combines multiple facts, constraints or reasoning steps on structured problems.
C3	Strategic	Abstract/causal reasoning, strategy construction and transfer to unfamiliar tasks.
C4	Expert	Robust reasoning through difficult, ambiguous, multi-constraint expert problems.
C5	Discovery	Derives new solution methods or explanatory models from evidence.
C6	Frontier-Generalizing	Integrates several domains; solves interconnected mission-scale cognitive problems.
CΩ	Open-Ended	Expands the representations and problem classes it can cognitively reach.

4.1 GUI Perception and Computer-Screen Understanding

Computer-screen understanding is a first-class capability for practical agents and is not a new top-level family. The framework treats GUI Perception as a diagnostic subscale of Cognitive Intelligence, $GP \subset C$. The containment matters in both directions: a model may reason well in text and fail to localise an interface element, and a model may read a screen correctly and still fail the task because planning, action selection, recovery or verification is weak. Collapsing the two makes computer-agent failures illegible.

GUI perception is broader than OCR. A screen-reading system must recognise text and icons, identify interface objects, ground them spatially, interpret interface state, track change across frames, and generalise to layouts it has not seen.

A computer-agent episode decomposes into $[P,G,S,A,R]$: perception, grounding, interface-state understanding, action selection, and result recognition. P, G and S diagnose C through GP; A and the closed loop are evaluated inside DI; R also touches SA, because the system must tell an intended success from a stale or unexpected screen. The decomposition is diagnostic and must not be counted twice in HLIS.

The observation channel is part of the measurement

ScreenSpot-Pro [13] and its predecessors give grounding evidence; OSWorld [28] gives end-to-end executable evidence. Neither number means anything without the harness beside it.
Benchmark version, observation channel (raw screenshot, OCR, accessibility tree, set-of-mark, or a combination), screen resolution and action interface must all be frozen and reported. These are harness

GP	Capability	Operational criterion
GP0	Pixel and text recognition	Recognises visible text, basic icons, colours and simple visual elements, claiming no task understanding.
GP1	UI-element recognition	Distinguishes buttons, menus, fields, tabs, dialogs, lists and windows as actionable objects.
GP2	Spatial grounding	Maps a referred element to the correct region, and represents containment, adjacency, ownership and target relations.
GP3	Interface-state understanding	Identifies enabled/disabled, selected, focused, open/closed, loading, error and modal states correctly.
GP4	Dynamic GUI understanding	Tracks change across screenshots and actions: what changed, whether the intended effect occurred, what is unresolved.
GP5	Novel-interface generalisation	Grounds and interprets unfamiliar software without application-specific scripting or a memorised path.

choices, they change performance materially, and a GP or DI figure quoted without them is a figure about an unnamed system.

5 Individual and Organizational Intelligence

5.1 Individual Intelligence (I)

Individual Intelligence concerns what a persistent individual can preserve, learn, and change about itself over time. It is not identical to base-model quality: a fixed but brilliant model can be high-C and low-I, while a persistent learning system can be moderate-C and higher-I.

5.1.1 Memory Capability M is an independent supporting coordinate

Memory is still not a sixth conceptual family: memory capacity by itself is not intelligence. But it is no longer defined as a relabelling of the I ladder either. Versions up to 1.7 bound memory stages one-to-one to I-levels, which makes a real and common profile inexpressible — a system with elaborate durable memory and little adaptive learning. **M is therefore measured independently, on its own benchmark, and reported beside I rather than inside it.**

The dependency is one-way

For a system A, let $M(A)$ be its highest independently certified memory level and $I(A)$ its highest certified Individual Intelligence level. Then

$$I(A) \geq I_n \implies M(A) \geq \mu_n,$$

and the converse is explicitly false. Memory is necessary for some forms of Individual Intelligence and is never sufficient for them. An I-level may not be promoted because M is high.
M stays outside the default HLIS product, because I already carries the learning and persistence claims and scoring both would count the same evidence twice.

Long context and long-term memory remain separate, and the separation now has two consequences rather than one. Long context makes more information available *inside* one inference episode. Long-term memory preserves information *across* episodes, restarts and elapsed time; retrieves it with provenance;

distinguishes current from obsolete state; and consolidates when consolidation is claimed. RULER- and LongBench-style evidence contributes to the context component of C, and certifies neither M1 nor I1: nothing in it survives a restart.

M	Independent memory capability	Representative certification evidence
M0	Ephemeral: active working context only; no durable state required after the episode.	Within-episode state use, and no unsupported claim of cross-session persistence.
M1	Persistent: durable storage and retrieval across sessions and restarts.	A real process restart followed by recovery of explicitly stored state.
M2	Structured episodic: provenance-aware episodes with temporal order, relevance selection, and current-versus-obsolete distinction.	Retrieval precision and recall under distractors; provenance accuracy; stale-state suppression.
M3	Consolidating: episodes become semantic knowledge and reusable procedures; contradiction, update and demotion are handled.	Held-out transfer attributable to consolidation; proceduralisation; contradiction update.
M4	Self-managing: confidence, utility, conflict resolution, repair, pruning, snapshots and recovery policies.	Memory-health detection; repair and pruning; recovery after deliberate corruption; snapshot lineage.
M5	Longitudinal knowledge system: hypotheses, experiments, evidence, decisions, failures and change lineage across projects.	Research and decision lineage; cross-project retrieval; evidence-graph consistency; reproducibility.
MΩ	Evolving memory architecture: representation, retrieval, consolidation or management mechanisms change under frozen external evaluation.	Candidate mechanism changes evaluated on hidden retention and transfer suites, with rollback and lineage.

5.1.2 Memory prerequisites for Individual Intelligence

I and M are evaluated separately and then joined by a gate. The thresholds below are working definitions requiring calibration, and they deliberately permit profiles such as I1/M4.

An I_n claim is invalid unless $M \geq \mu_n$, and the gate is checked separately from the I evidence. This is what stops an elaborate memory subsystem from being read as learning.

Each memory requirement is a test, not a component list. An I1 claim is established by suspending the system, restarting the process, and observing that it recovers task state and can say where that state came from — not by pointing at a vector database. An I2 claim requires showing that a lesson drawn from one episode changes behavior in a later one and that the change survives when the retrieval index is rebuilt. The distinction matters because storage is easy to install and consolidation is not, and a framework that accepted the presence of a store as evidence of the function would grade the installation rather than the individual.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on role differentiation, evidence-bearing communication, proposer/evaluator/promoter separation, persistent organizational state, and the ability to learn or redesign the organization itself.

6 Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be very difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

I	I-specific cumulative capability	Minimum μ_n
I0	Transient, reactive individual operation.	M0
I1	Persistent individual continuity and reliable task and identity resumption.	$\geq$ M1
I2	Verified experience changes later behavior while the learning mechanism stays fixed.	$\geq$ M3
I3	Governed modification of cognitive mechanisms under independent evaluation.	$\geq$ M4
I4	The improvement process itself improves under external evaluation.	$\geq$ M4
I5	Unknowns are converted into independently validated knowledge.	$\geq$ M5
IΩ	Open-ended individual evolution expands cognitive instruments and problem reach while retaining validation lineage.	$\geq$ M5; MΩ only when evolution of the memory architecture is itself part of the claim

$$S_A(T,H) = P(\text{success} \mid \text{task difficulty } T,\ \text{human cognitive intervention budget } H) \tag{1}$$

$$F_A(H,p) = \max\{T : S_A(T,H) \geq p\} \tag{2}$$

Version 1.4 note

Equation (1) is the Version 1.3 primitive. §13.9 replaces it for scoring purposes with a delivered-outcome primitive $S_A^{\text{net}}(T,H)$; the frontier $F_A(H,p)$ is unchanged in form and is computed from the net surface.

6.1 Task Difficulty (T)

6.2 Human Cognitive Intervention (H)

Because lower H means less required human cognition, unified gates use $H \leq H_n$ rather than $H \geq H_n$. Governance interventions are logged separately: an approval to deploy does not demonstrate a cognitive deficiency unless the approval also supplies task strategy.

O level	New cumulative capability
O0	Coordination/routing among agents.
O1	Persistent organizational memory, role registry, task/evidence history.
O2	Adaptive routing, role assignment or resource allocation based on evidence.
O3	Self-improving organization: validated modification of organizational architecture and workflows.
O4	Recursive organizational improvement: improvement of the mechanism that discovers organizational improvements.
O5	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
OΩ	Open-ended organization: discoveries create new agent types, roles, tools and organizations that expand collective reach.

T	Band	Definition
T0	Atomic	Single obvious operation; procedure explicit.
T1	Routine	Short familiar task; goal clear; agent infers a small procedure.
T2	Multi-step	Several dependent steps with a clear outcome and verifier.
T3	Complex	Strategy choice, uncertainty, recovery, replanning or diverse tool use.
T4	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
T5	Frontier	Solution method initially unknown; discovery or invention required.
T6	Mission	Human supplies a mission; the system generates projects/tasks and manages changing conditions.
TΩ	Open-ended charter	Human supplies a bounded charter; the system repeatedly identifies valuable missions and expands future reach.

6.3 Reliability p is a certification condition, not a seventh axis

One successful run is insufficient. Claims should be expressed at a reliability threshold, for example $F_A(H1, 0.80) = T4$. Reliability determines confidence in the demonstrated frontier but does not become another conceptual intelligence family.

7 Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures the evidence-grounded ability of a system to represent, monitor, predict and reason about its own state, capabilities, limits, authority, failures, modifications, role and identity lineage.

8 Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained.

$$U_n \iff C \geq C_n \wedge I \geq I_n \wedge O \geq O_n \wedge SA \geq SA_n \wedge S_A(T_n, H_n) \geq p_n \tag{3}$$

$$\text{Retention condition: } U_n \Rightarrow U_{n-1}$$

H	Intervention budget	Meaning
H0	None	No human cognitive assistance during execution. Governance approval can remain separate.
H1	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
H2	Occasional	Bounded clarification or local correction; no central strategy.
H3	Periodic	Periodic review and moderate local guidance.
H4	Frequent	Repeated cognitive guidance and next-step correction.
H5	Continuous	Human supplies most or all major steps.

SA	Name	Operational criterion
SA0	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
SA1	State awareness	Knows identity, role, current task/phase, basic resources and status from grounded state.
SA2	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information and freshness.
SA3	Causal self-awareness	Diagnoses why its own reasoning or action succeeded or failed and identifies implicated mechanisms.
SA4	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions and compares predictions to observed effects.
SA5	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
SA6	Mission/role awareness	Maintains a coherent self-model across projects, roles, responsibilities and organizational relationships.
SAΩ	Reflexive open-ended	Maintains an evidence- and provenance-linked self-model across continued cognitive or objective evolution.

9 Why the Unified Level Should Not Be an Average

Consider a system with $[C5, I4, O2, T4, H1, SA4]$. Averaging encoded numbers might produce a score near level four, yet organizational intelligence remains only O2. If the system is being certified as an organizational intelligence, O2 is a structural bottleneck and the system cannot validly claim U4. The gated rule reports U2 with an explicit profile showing which dimensions are already ahead of the bottleneck.

Example report: `U2 [C5, I4, O2, T4/H1, SA4] -- bottleneck:  O2`

The word *better* must be interpreted carefully. U5 is better than U4 in this framework because it has a broader validated capability, autonomy and evolution envelope. A specialized U1 tool may still be faster, cheaper, safer or more accurate on a narrow task. Unified level is a dominance relation over required capabilities, not a universal performance ordering.

10 Singleton and Organizational Certification

Organizational intelligence should not artificially cap an individual that is not an organization.

$$UI_n \iff C \geq C_n \wedge I \geq I_n \wedge SA \geq SA_n \wedge S_A(T_n, H_n) \geq p_n \quad (4)$$

$$UO_n \iff UI_n \wedge O \geq O_n \wedge \text{organization-level delegation and self-awareness evidence} \quad (5)$$

U	C	I	O	Delegation gate	SA	Integrated meaning
U0 Reactive	C0	I0	O0	T0 at H5–H4	SA0	Executes explicit instructions.
U1 Persistent	C1	I1	O1	T1 at ≤H3	SA1	Maintains state; completes small delegated goals.
U2 Adaptive	C2	I2	O2	T2 at ≤H2	SA2	Learns from experience; completes persistent multi-step work.
U3 Self-Improving	C3	I3	O3	T3 at ≤H1	SA3	Chooses strategy, diagnoses itself, improves methods.
U4 Recursive	C4	I4	O4	T4 at ≤H1	SA4	Recursively improves improvement; manages long-horizon projects.
U5 Discovery	C5	I5	O5	T5 at ≤H1	SA5	Discovers unknown solution methods and validates them.
U6 Mission	≥C5	≥I5	≥O5	T6 at ≤H1	SA6	Turns mission into goals, projects, tasks, execution and evaluation.
UΩ Open-Ended	CΩ	IΩ	OΩ	TΩ at H0	SAΩ	Identifies worthwhile missions; discoveries expand future intelligence.

For a multi-agent organization, member capabilities are reported alongside *O* rather than collapsed into a fictitious "average individual." Organizational intelligence can arise from heterogeneous roles, and lower self-write roles can be essential as independent evaluators or promoters.

11 Harness Realization of the Unified Hierarchy

The levels-of-automation tradition [5, 19, 25] established that autonomy is graded and multi-stage; Klein et al. [11] and Amershi et al. [1] add the requirement that automation acting as a team member be predictable and directable; and Lee and See [12] supplies the account of reliance this framework's intervention axis operationalizes. The framework is designed to be realized by a harness. The LLM supplies generative cognition; the harness supplies persistence, permissions, evidence flow, learning, self-modification gates, organizational boundaries, delegation, external verification and self-modeling. The same base LLM can therefore instantiate different intelligence profiles depending on the architecture around it.

Harness generation is an engineering milestone, not a certified intelligence level. HG3 may contain the mechanisms required for I3 but still benchmark only at I2 or T2. **The architecture enables a claim; the benchmark earns it.**

11.1 The Harness-LLM Pair Is the Basic Experimental Unit

Neither factor has a level on its own. A benchmark result is a property of the pair $A = A(m,h)$, and a number attributed to the model alone attributes a joint product to one of its factors. Version 1.4 states this more strongly than earlier versions because it is now measured rather than argued: §17.4 reports a case in which the same frozen model moves across harness generations, and §14.3 isolates the part of that movement attributable to the model.

11.2 Level-by-Level Harness Realization

11.3 The Ladder as Built (new in 1.4)

The table above states *required structure*. A measurement needs a specific instantiation, and Version 1.4 records one so that reported curves are comparable. The reference ladder used in §17.4 instantiates the first

Harness generation	Mechanisms added	Primary levels enabled
HG0	LLM, ephemeral context, bounded tools, evidence log	C measurement, I0, basic T0–T1 work
HG1	Persistent episodic/semantic/task/self state, retrieval, exact replay	I1, SA1–SA2, stronger T1–T2 delegation
HG2	Evidence-backed consolidation and tested procedures	I2, adaptive autonomy, stronger T2–T3
HG3	Governed candidate modification of cognitive mechanisms plus frozen evaluation	I3, SA3–SA4
HG4	Meta-improvement engine whose own proposal process can be externally improved	I4
HG5	Unknown/hypothesis/experiment/knowledge loop	I5, SA5, T5 frontier tasks
HG6	Persistent organization, role/authority separation, mission manager	O1–O5 development, SA6, T6 mission autonomy
HGΩ	Validated discovery-to-new-instrument/agent/organization transduction	IΩ/OΩ/SAΩ pathway, TΩ charter autonomy

four rungs from delegation-harness mechanisms, under one rule: **each rung is a strict superset of the one below**. Without that, a difference between rungs is attributable to nothing.

Why HG0 has no acceptance step

A harness that cannot accept cannot report success either: everything it produces is asserted. HG0's score is therefore computed by the benchmark's verifier from outside. That makes HG0 the honest baseline, because it is the configuration a contemporary leaderboard measures.

11.4 A Base LLM Can Move Up the Ladder Through Better Harness Design

A fixed LLM need not have a fixed agent intelligence level. If HG0 exposes only ephemeral prompting, the same model may behave like a low-persistence tool; HG1–HG3 can unlock memory, planning, delegation, learning, self-modeling and governed self-improvement.

The LLM may also design candidate harness improvements. Such proposals must be implemented in a sandbox and evaluated with unchanged hidden benchmarks, authority rules, resource budgets and external verifiers. Define **Harness Design Gain** as $HDG(m) = HLIS(m, h_{\text{candidate}}) - HLIS(m, h_{\text{baseline}})$. A positive HDG demonstrates harness-design competence.

The self-certification rule (stated explicitly in 1.4)

A model that designs the harness which certifies it has placed the acceptance criterion inside its own write set, and capability makes this worse rather than better, since a more capable designer is better at finding the arrangement that passes.

The designer may design. It may not author, modify or select the tasks, the verifiers, or the acceptance criteria used to certify the result. **The certification set is frozen before the harness is designed and held by a locus the designing model cannot write to.**

Target	Minimum harness structure around the LLM	What the LLM must do	Matched certification dataset
U0 / HG0	Ephemeral context; bounded tools; action log; independent result check.	Follow explicit instructions and produce externally checkable outputs.	U0/DL0 atomic tasks; exact-match or deterministic tool/test oracle.
U1 / HG1	HG0 + persistent $E/S/I/Q$ state, recent context, checkpoints, scoped retrieval, replay.	Recover identity/task state, continue after restart, infer a short routine procedure.	U0 retention + U1 persistence/restart tasks + T1 delegation tasks.
U2 / HG2	HG1 + evidence-linked consolidation, semantic promotion, verified procedures, fixed learning rule.	Learn from episodes and reduce repeat errors without changing the learning mechanism.	U0–U1 retention + I2 learning transfer + T2 multi-step tasks under bounded H.
U3 / HG3	HG2 + candidate Θ modification lane, frozen replay/hidden tests, independent evaluator and promoter; multi-agent O3 structure if organizational.	Diagnose internal failure, propose a better cognitive/organizational mechanism, choose strategy for T3 tasks.	U0–U2 retention + I3/O3 self-improvement + T3/H1 delegation + SA3 causal self-model tests.
U4 / HG4	HG3 + explicit improvement engine Ψ, longitudinal change ledger, meta-improvement benchmark, long-horizon project manager.	Improve the process that finds improvements while retaining earlier abilities; manage difficult T4 projects.	U0–U3 retention + recursive-improvement competence + T4 long-horizon tasks + SA4 change-prediction tests.
U5 / HG5	HG4 + unknown, hypothesis, experiment and knowledge state, experiment selector, novelty environment, discovery verifier.	Discover previously unknown mechanisms or solution methods and validate them on sealed cases.	U0–U4 retention + hidden-mechanism/discovery datasets + T5 tasks + SA5 self-unknown experiments.
U6 / HG6	HG5 + mission manager, goal/project/task generator, portfolio/resource state, dynamic-environment replanning.	Turn a mission into projects/tasks, reprioritize under change, meet mission-level criteria with little human cognition.	U0–U5 retention + dynamic mission environments + T6/H1–H0 tests + SA6 role/mission-state tests.
UΩ / HGΩ	HG6 + bounded charter, opportunity model, knowledge-to-instrument transduction, new-agent/role proposals, frozen external governance.	Generate useful missions, convert validated discoveries into reusable capabilities, expand the reachable problem frontier.	All retention suites + repeated charter-to-mission cycles + frontier-expansion and lineage tests.

12 A Three-Stage Evaluation Strategy

12.1 Stage A — Delegation Intelligence as the First Operational Screen

The first benchmark should be Delegation Intelligence because it directly answers the most operational question: what difficulty of task can be handed to this system, at what verified success rate, and with how much human cognitive intervention? For every LLM-harness pair estimate $S_A(T,H)$ and report $F_A(H,p)$. This stage is inexpensive relative to full self-improvement or organizational evaluation and quickly distinguishes a strong model in a weak harness from a genuinely autonomous agent system.

12.2 Stage B — Unified Intelligence Certification

After a pair has a stable delegation profile, evaluate the remaining core requirements: C, I, O when applicable, and SA. Unified promotion remains gated: a pair earns U_n only when it satisfies the level-n requirements and all lower-level retention suites.

Rung	Mechanisms added	Attempts	Acceptance	Reversible	Routes
HG0	the model, bounded tools, an evidence log	1	**none**	no	no
HG1	persistent state; a criterion registered before the work exists; acceptance in a separate process	1	yes	no	no
HG2	snapshot before the first mutation; restore on rejection; retry with the failed check named	3	yes	yes	no
HG3	failure classification by kind; evidence-based competence model; routing across executors	4	yes	yes	yes

Table 1: The reference ladder `dli-ladder/HG0-HG3`. A curve is comparable only against the same ladder identifier.

12.3 Stage C — Application-Specific Coordinate Panels

Real applications need not weight every coordinate equally. For a persistent scientific or coding worker, I may be central; for a multi-agent laboratory or software fleet, O; for high-autonomy systems, SA and verification strength; for model research, C; for embodied or infrastructure systems, Circle completion λ may be added as an orthogonal application coordinate. Application panels refine the profile without changing the common U certification rule.

12.4 Every Coordinate Level Requires a Harness Contract and a Dataset

Coordinate family	Harness contract	Dataset family
Delegation $DI = (T, H, p)$	Task intake, planner, state, tools, intervention logger, independent verifier.	DLI-Bench: T0–TΩ tasks crossed with H0–H5 budgets.
Cognitive C	Minimal standardized context/tools so the test emphasizes problem solving rather than persistence.	C-Bench: broad reasoning, math, language, visual/spatial or domain batteries.
Individual I	Persistent identity/state, memory, learning, then governed self-/meta-improvement machinery.	I-Bench: restart, transfer learning, Θ-change, Ψ-change, discovery lineage.
Organizational O	Multiple real agent processes/roles, shared evidence, routing, separation, promoter/evaluator boundaries.	O-Bench: coordination, adaptive routing, reorganization, recursive organization, collective discovery.
Self-awareness SA	Evidence-grounded self-model with state freshness, capability/authority representation, change lineage.	SA-Bench: state truth, limitation calibration, failure attribution, change prediction, identity lineage.
Circle / λ	Substrate adapter with observe-propose-implement-validate-deploy-measure contract.	Circle-Bench: substrate-specific closure tests.

13 Overall Scoring for One Harness-LLM Pair

13.1 Keep the Ordinal U-Level and Add a Continuous Pair Score

The highest certified Unified level U^* remains the primary ordinal statement. A continuous score is useful for comparing systems within the same level, measuring near-promotion progress and quantifying whether a harness change helped. For a frozen pair of base model m and harness h:

$$HLIS(m,h) = 100 \times \left[\prod_{d \in D} A_d^{\alpha_d} \right]^{1/\sum_d \alpha_d} \tag{6}$$

13.2 Formal Definition and Meaning of the Symbols

m is a frozen base model with a stated version and hash. h is a frozen harness with a stated manifest. D is the declared dimension set, $\{C, I, DI, SA\}$ for singleton agents and $\{C, I, O, DI, SA\}$ for organizations. $A_d \in [0,1]$ is a cumulative achievement variable for dimension d that already enforces lower-level retention. $\alpha_d > 0$ are predeclared weights. The complete notation is $HLIS(m,h;D,\alpha,R,B)$, making the resource envelope R and benchmark version B explicit.

13.3 Choosing the Dimension Set D Without Double Counting

A dimension enters D once. Evidence that contributes to A_C does not also contribute to A_{DI} merely because the task was difficult; a benchmark contributes to the coordinate it is designed to isolate.

13.4 Constructing the Achievement Variables

Each A_d is normalized within declared families before aggregation, so that a domain with many cheap items cannot dominate a domain represented by fewer but harder tasks.

13.5 Why HLIS Uses a Weighted Geometric Mean

A simple arithmetic mean allows an exceptional score in one dimension to compensate too easily for a severe weakness in another.

Profile	Arithmetic mean	Geometric mean	Interpretation
$[0.80, 0.80, 0.80, 0.80, 0.80]$	0.80	0.80	Balanced capability.
$[1.00, 1.00, 1.00, 1.00, 0.10]$	0.82	≈0.63	One severe weakness is visibly penalized.

What the geometric mean does and does not do (clarified in 1.4)

It supplies *partial* bottleneck sensitivity, not full. On the profile of §9, raising a coordinate already three levels past the bottleneck still moves HLIS by several points while the coordinate blocking promotion does not move at all.

That is acceptable only because HLIS is explicitly **not** the promotion rule. The hard ordinal gate does the bottleneck work; HLIS orders systems within a gate. Reporting HLIS without U^* and without the named bottleneck reintroduces exactly the concealment the gate exists to prevent, and Version 1.4 therefore makes both mandatory in the reporting standard (§13.15).

Implementations that want the continuous score to carry the gate's property may report the optional **bottleneck-margin** form: $U_n^* + m$, where $m \in [0,1)$ is the fraction of the distance to gate $n+1$ covered by the bottleneck coordinate *alone*. Every other coordinate contributes exactly zero. It is sortable and cannot conceal; it discards information about how far ahead the other coordinates are, which the profile carries anyway.

13.6 The Weight Parameters α_d

For a general leaderboard, equal α_d is recommended because it minimizes hidden value judgments. Application-specific secondary scores may use different weights, but the weights must be declared before the hidden benchmark is run and never adjusted after observing a result.

13.7 Computing Cognitive Achievement A_C

$$A_C = G(R_{\text{general}}, R_{\text{science}}, R_{\text{math}}, R_{\text{abstract}}, R_{\text{code}}, R_{\text{multi}}, R_{\text{long}}) \tag{7}$$

Contemporary benchmarks such as HLE [22], GPQA [23], FrontierMath [6], ARC-AGI [2], LiveCodeBench [9], MMMU [30] and RULER [8] contribute evidence, with exact versions and normalization rules published.

13.8 Cumulative Achievement for I, O and SA

Let $q_{d,k}$ be externally verified performance on the level-k suite for dimension d. To prevent level skipping:

$$q^*_{d,k} = \min_{j \le k} q_{d,j}, \qquad A_d = \frac{\sum_k w_k q^*_{d,k}}{\sum_k w_k}, \quad d \in \{I, O, SA\} \tag{8}$$

A system cannot receive full continuous credit for I4-like behavior if it fails an I2 retention requirement.

13.9 Delegation Achievement A_{DI} — revised in Version 1.4

13.9.1 The Version 1.3 definition, and what measurement showed

Version 1.3 defined the primitive as $S_A(T,H) = P(\text{success})$ and the achievement variable as its weighted mean:

$$A_{DI}^{\text{v1.3}} = \frac{\sum_{t,h} w_t v_h S_A(T_t, H_h)}{\sum_{t,h} w_t v_h} \tag{9}$$

Equation (9) has a property that was not visible until the ladder was built.

Proposition 1 (Acceptance invariance)**.** *Let h and h' be two harnesses differing only in that h' adds an acceptance step: an independently applied criterion that may decline to accept a result. Then $S_A^h = S_A^{h'}$ on matched tasks, and therefore $A_{DI}^{v1.3}(h) = A_{DI}^{v1.3}(h')$.*

Argument. S_A is the probability that the system *succeeds*. An acceptance step changes neither what the system produces nor whether it is correct; it changes which results are reported as successes. A statistic defined on $P(\text{success})$ is therefore invariant to it. $\square$

Why this matters more than it first appears

The invariant rung is the one on which the reportability of every rung above it depends. This framework's own protocol (§18) requires an independent verifier throughout, precisely because a result accepted by its own producer is an assertion rather than a completed task.

It is also worse than blind. A harness that accepts everything scores at least as well as one that declines wrong work: strictness is invisible when correct and penalized whenever it declines something that would have passed. **The score gradient points away from the acceptance mechanism at every point.**

13.9.2 The Version 1.4 primitive: delivered outcomes

The repair is to define the primitive on what was *delivered* rather than on what succeeded. Four outcomes are distinguished, and only the two in bold are failures of the same kind:

Harness accepted?	Verifier passed?	Outcome
n/a (no acceptance step) or yes	yes	delivered and correct
n/a or yes	no	**false completion** — handed back as done, and wrong
no	no	held back — a correct refusal; costs human load, not reliability
no	yes	false rejection — the cost of strictness

$$S_A^{\text{net}}(T,H) = P(\text{verifier pass}) - \rho(\tau)\cdot P(\text{false completion}) \tag{10}$$

$$A_{DI} = \frac{\sum_{t,h} w_t v_h S_A^{\text{net}}(T_t,H_h)}{\sum_{t,h} w_t v_h} \tag{11}$$

where $\rho(\tau) = (C_{\text{detect}} + C_{\text{undo}} + C_{\text{residual}})/V$ is the class's loss ratio — the same quantity that gives the minimum admissible reliability $p^* = \rho/(1+\rho)$. A refusal appears nowhere in Equation (10): it is accounted for in human load, which §13.13 already requires beside the score.

Properties

It restores sensitivity to acceptance. Two harnesses differing only in an acceptance step now differ in A_{DI} by ρ times the false-completion rate they differ by.

It prices the two failures differently, as the delegating party does. A refusal costs someone's attention; a confident wrong answer costs whatever being wrong costs on that class, which ρ already names.

It is backward-comparable. Setting $\rho = 0$ recovers Equation (9) exactly, so a v1.3 figure can be reproduced and the difference audited rather than asserted.

It requires no new coordinate. Only the primitive changes.

One property to know about: the net surface floors at zero

S^{net} is clamped at zero, so a band whose pass rate is at or below ρ times its false-completion rate scores zero under both primitives and cannot distinguish them. At $\rho = 1$ that is any band where every failure was delivered.

This is the correct reading — a class on which everything wrong was handed back as finished has delivered nothing of value — but it means the measure saturates at the bottom, and a comparison between two harnesses is uninformative on bands where both are already floored. Report the per-band surface alongside the aggregate so a floored band is visible as such rather than as agreement.

The task weights w_t increase with difficulty and the intervention weights v_h increase as required human cognitive intervention decreases. One illustrative monotone schedule is $H0$=6, $H1$=5, $H2$=4, $H3$=3, $H4$=2, $H5$=1, with w_t doubling per T band. The exact schedule is a calibration choice and must be predeclared.

The scalar A_{DI} must never replace the frontier. At minimum report $F_A(H0, p)$, $F_A(H1, p)$ and $F_A(H2, p)$, because two agents can have similar surface averages while one reaches harder tasks only with frequent human help and the other is weaker but much more autonomous.

13.10 Numerical Example of HLIS

With equal weights and $A_C = 0.90$, $A_I = 0.70$, $A_O = 0.60$, $A_{DI} = 0.80$, $A_{SA} = 0.75$:

$$HLIS = 100 \times (0.90 \times 0.70 \times 0.60 \times 0.80 \times 0.75)^{1/5} \approx 74.3$$

The recommended report is not "HLIS=74.3" but `U3 | HLIS 74.3 | [C5,I3,O3,T4/H1,SA3] | p=0.80 | bottleneck O3 | Bv1.4 | R-Standard`.

13.11 Zero Scores, N/A Dimensions and Hard Gates

A zero in a required geometric-mean component forces HLIS to zero. This must not be confused with "not applicable": for a singleton system, O is **omitted from** D, not set to zero. A scoring implementation must name the omitted dimensions in its output, because a score over a subset is not comparable with one over the full set.

13.12 The Run Log Must Record Acceptance (new in 1.4)

Proposition 1 cannot be detected, let alone repaired, unless the underlying records distinguish the outcomes. A run log that stores only `verifier_pass` gives an identical row to a run that declined wrong work and a run that handed it over.

Required run-log fields

`verifier_pass` (external verdict) and `harness_accepted` (did the harness accept; null where the harness has no acceptance step), from which `false_completion`, `held_back` and `false_rejection` are derived. Also `attempts`, and `verification_responsive` for §14.3.

13.13 Reliability, Confidence Intervals and Resource Envelopes

Every achievement estimate is sampled and therefore uncertain; HLIS should carry a confidence interval. Resource conditions must accompany the score: a 500-call multi-agent harness and a one-call harness are not automatically comparable. Recommended R fields include total tokens, LLM calls, wall-clock time, compute class, persistent-memory budget, tool access, external knowledge access and subagent count. Human load — intervention count, human cognitive minutes and authorization latency — is reported here rather than inside the intelligence score.

13.14 HLIS Is the Building Block of HIL

$$HSC_m(k) = HLIS(m, HG_k) \tag{12}$$

13.15 Recommended Pair-Level Reporting Standard

A pair report is incomplete unless it carries: U^*; HLIS with its dimension set and weights; the discrete profile; the named bottleneck; reliability with an interval; the frontier at H0, H1 and H2; the false-completion rate; the resource envelope; and the benchmark and ladder versions.

Added in Version 1.6. Two further entries, both concerning the instrument rather than the system:

- **Per-coordinate headroom** — the fraction of items in the coordinate's suite on which the strongest available executor is *not* already at the maximum score. A coordinate whose headroom is near zero is reporting its suite's ceiling, and its contribution to any composite must carry that statement (§16.6).
- **Concordance-audit status** — whether the suite's items have been audited for concordant cross-tier error since the last change to any answer key, and how many items the audit flagged (§16.3).

14 Harness-Intelligence-Level (HIL) Characterization of a Base LLM

14.1 Why an LLM Needs a Harness-Based Score

A base LLM should not be judged only in a minimal chat harness, because some models are unusually capable of using memory, planning, tools, multiple roles, verification and self-improvement machinery. Conversely, an elaborate harness can mask a model that contributes little. Evaluate the same frozen model across a standardized ladder $\mathscr{H} = \{HG0, \ldots, HG6, HG\Omega\}$, with tools, external knowledge access, resource ceilings, authority and hidden benchmarks standardized, so the comparison reflects model-harness interaction rather than uncontrolled infrastructure advantages.

14.2 HIL-Level, HIL-AUC, HIL-Ceiling and Harness Gain

Metric	Definition	Interpretation
HIL-Level(m)	Highest standardized generation k for which $A(m, HG_k)$ satisfies the cumulative target gate.	How far up the ladder the LLM can validly exploit the architecture.
HIL-AUC(m)	Weighted mean of $HLIS(m, HG_k)$ across tested generations.	Breadth across harness complexities. *See the caution in §14.4.*
HIL-Ceiling(m)	$\max_k HLIS(m, HG_k)$.	Best score a standardized harness realizes from the model.
Harness Gain(m)	$\max_k HLIS(m, HG_k) - HLIS(m, HG0)$.	Capability unlocked by harnessing beyond the minimal baseline.
Harness Design Score	Mean independently verified gain when the LLM proposes harness improvements under a fixed design budget.	Ability to improve the architecture that harnesses it.
$\rho_V(m)$ **(new in 1.4)**	See §14.3.	The part of Harness Gain attributable to the model.

14.3 Verification Responsiveness ρ_V (new in 1.4)

Harness Gain is a difference of two scores. It says how much the harnessing was worth and not why, so it cannot distinguish two models with the same gain arising from different mechanisms — and those two models should be built around differently.

A harness raises two things, and only one requires the model

Acceptance and reversibility remove false completions and unrecoverable actions. They work on a model that contributes nothing beyond its first attempt.
Retry, routing and replanning require the model to convert an independently detected failure into a correction. A model that cannot do this gains nothing from them.

Definition 1 (Verification responsiveness)**.** Over runs where the acceptor rejected attempt k,

$$\rho_V = \frac{\#\{\text{accepted at } k{+}1 \text{ with no help deeper than CID1}\}}{\#\{\text{rejected at } k\}}$$

ρ_V is cheap — it needs one extra attempt on rejected runs, not a whole ladder — it is defined on the model's behavior rather than on a score difference, and it predicts which part of Harness Gain a model can realize. It is invisible to any single-attempt evaluation, because it is defined on the second attempt.

A failure mode of ρ_V, found in measurement

ρ_V computed over *coarse* criteria understates the model. In one observed case a model failed a class twice because the rejection named a rule it had not broken — the criterion bundled two rules under one name. Splitting the criterion so the message named the actual failure, and changing nothing else, moved the class from 0/2 to 2/2.
The rejection must be accurate about *what* failed, not only *that* something did. ρ_V therefore measures the pair's feedback quality as well as the model's responsiveness, and criteria granularity must be reported with it.

14.4 A Single HIL Score, and Two Cautions

For applications requiring one 0–100 number:

$$\text{HIL-Score} = 0.55 \times \text{HIL-AUC} + 0.35 \times \text{HIL-Ceiling} + 0.10 \times \text{Harnessability} \tag{13}$$

where Harnessability is a 0–100 normalization of positive Harness Gain. These weights are a proposal rather than a law. The three components and HIL-Level must always be published beside the composite.

Two structural cautions (new in 1.4)

HIL-AUC and HIL-Ceiling are computed from the same series. Ceiling is the maximum of the values AUC averages, so their correlation is structural rather than empirical, and $0.55 + 0.35 = 0.90$ of the weight sits on two views of one quantity. The only component not derivable from the others carries 0.10.
HIL-AUC does not measure harnessability. Consider a flat model scoring 55 at every rung and a steep one scoring $20, 35, 55, 75, 92$. Their AUCs are 55.0 and 55.4 — essentially equal — though only one converts architectural support into verified intelligence, which is the property §14.1 says HIL exists to measure. The composite does separate them, through the two terms carrying least weight.
Until calibration settles this, **the curve itself is the primary result** and the composite is a convenience. Report $\langle$HIL-Level, $HLIS(m, HG0)$, HIL-Ceiling, Harness Gain, ladder id$\rangle$ — the ordinal statement plus the curve's start, top and rise.

Score	Subject being measured	Primary use
C-Score / model baseline	Base LLM in a minimal standardized harness.	Raw cognitive problem-solving comparison.
Delegation Surface / DI	One LLM-harness pair.	How hard a task can be delegated at each intervention budget.
Unified level U^*	One pair or organization.	Highest cumulative stage actually certified.
HLIS	One frozen pair.	Continuous comparison within and across levels.
HIL-Level + curve	One base LLM across the ladder.	How much intelligence the model expresses when harnessed.
ρ_V	The model, measured through a pair.	Whether the model converts detected failure into correction.
HDS / HDG	LLM as harness designer under frozen external evaluation.	Ability to design architectures that improve realized intelligence.

14.5 Recommended LLM Scorecard

15 Benchmark Library for Harness-Level Intelligence

A harness intelligence program should maintain a versioned benchmark registry rather than one monolithic test set. The minimal unit is a task manifest containing target coordinate/level, frozen model and harness hashes, tool/resource/authority budget, intervention policy, hidden verifier, success threshold and required retention suites.

Suite	Purpose	Example level-specific evidence
DLI-Bench	First-stage delegation surface.	T0 atomic; T2 multi-step persistence; T3 strategy construction; T4 long-horizon project; T5 hidden solution; T6 mission; TΩ charter cycles, each under H budgets.
U-Bench-U0…UΩ	Unified promotion and lower-level retention.	At U_n, rerun $U_0 \dots U_{n-1}$ retention plus the new gate.
I-Bench	Persistent individual evolution.	Restart continuity, learning transfer, Θ modification, Ψ improvement, discovery, lineage.
O-Bench	Collective intelligence.	Routing, persistent organization, adaptive coordination, reorganization, collective discovery, new-role creation.
SA-Bench	Evidence-grounded self-awareness.	State truth, capability limits, causal self-diagnosis, predicted self-change, self-unknowns, identity lineage.
C-Bench	Broad cognitive capability.	Multiple reasoning/domain families, to avoid mistaking one specialty for general cognition.
Circle-Bench	Substrate reach, reported orthogonally.	Substrate-specific implementation and validation loops.

15.1 Specification Status Must Be Recorded Per Task (new in 1.4)

A registry of task *specifications* is not a benchmark, and the difference must be visible per row rather than in aggregate. Version 1.4 requires three statuses:

- **bound** — a runnable instance exists for this band *and* this family;
- **band-only** — something runs at this difficulty band but not for this family;
- **specification-only** — nothing runs it.

Band-only must not be counted as coverage. Substituting a software-engineering task for a document-workflow task at the same band measures a different task, and counting it reports coverage the suite does not have.

The reference library at the time of writing binds 34 of 224 specified benchmarks, marks 30 band-only and leaves 160 as specifications — a state worth reporting plainly, because a library that claims it can run something it cannot is worse than one that claims nothing.

16 The Validity of the Items (new in 1.6)

Everything in §13 and §14 is arithmetic performed on the outcomes of benchmark items. The arithmetic has been examined closely — Version 1.4 found a structural invariance in it and repaired it. The items have not. An item with a withheld answer key is an assertion by its author that a particular response is correct, and until something tests that assertion the whole score inherits its errors.

This section reports what happened when the delegation suite was pushed for headroom, and turns the results into three requirements.

16.1 Trickiness does not desaturate a suite; difficulty does

Version 1.4 reported saturation: frontier executors had little headroom, and two rungs that differed materially produced identical aggregate curves. The first attempt to fix this built three items around specific reasoning hazards — an operation whose result depends on the order two events are applied in, a daily aggregation spanning a daylight-saving transition, and an identity comparison over Unicode strings that normalise to the same form.

They worked as traps and failed as measurements.

Item family	Scripted-correct solver	Naive solver	Live executors
Three DL3 hazard items	10/10	0/10	all probes passed, both tiers

The items separate a correct implementation from a careless one perfectly and separate no model at all. The reason is visible once stated: an item constructed around a hazard has to describe the hazard in order to be answerable, and a competent reader who has been told that the events may arrive out of order will handle events arriving out of order. The hazard is in the task text, so reading the task text is sufficient.

On what makes a delegation item discriminative

A hazard an item must disclose to be well-posed is not a source of difficulty. It is a reading-comprehension check that every capable executor passes.

Difficulty that survives disclosure comes from *quantity* — a specification with more interacting rules than can be held at once — or from *search* — a question whose answer requires establishing that no better answer exists. Both can be stated completely in the open.

16.2 Two harder item families

Two DL4 families were built on those two axes. Neither conceals anything: both publish a complete specification, and what is withheld is only which instances are used to score.

Specification scale. A small expression language with ten interacting rules — short-circuit operators that return an operand rather than a boolean, four-way truthiness, typed arithmetic with an error value that propagates, a binding form looser than every operator, and a precedence table — scored against roughly 238 generated expressions per seed. Four dialects vary the rounding direction of integer division and the treatment of cross-type equality, so an implementation memorised from one instance is wrong on the next.

Search. A minimum-cost covering problem: cover each day of a schedule to its stated demand by selecting from a set of costed shift patterns, at minimum total cost. Sixty instances per seed, with two variants over whether a pattern may be selected more than once. A selection scores only if it is both feasible and exactly optimal, where the optimum is computed by exhaustive search over the coverage state space at build time.

The second family separates two failure modes that a boolean verdict reports identically. Measured against its own reference solvers across four seeds:

Solver	Feasible	Optimal	Reading
Exact search	60/60	60/60	correct
Cost-per-covered-day greedy	60/60	3–12/60	always a valid schedule, rarely the cheapest

The greedy solver is the plausible wrong answer, and it is wrong in a way that a feasibility check cannot see: it never returns an invalid schedule. Instance draws are filtered at build time by replaying both solvers, so that the obvious method is verified to be suboptimal on most instances rather than assumed to be — on the four seeds measured, on 48 to 57 of 60.

16.3 Cross-tier concordance auditing

The expression-language family was then run against two executors of deliberately different capability. The purpose was to see whether a scale-difficulty item discriminates. It did something more useful first.

Seed	Frontier executor	Weaker executor	Status
0	0.941	0.933	*both wrong on the same case*
1	0.958	0.958	*both wrong on the same case*
2	1.000	0.996	
3	1.000	1.000	

Read as a measurement, this is a discriminating item: two tiers, different scores, graded rather than binary. Read as evidence about the item, the first two rows say something else. The two executors did not merely both fail; they returned the *same value* for the same expression, and the key disagreed with both.

Both cases were defects in the item.

The two defects together accounted for 14 of 238 cases for one executor on the first seed and 16 for the other; how the count divides between them was not established, and after both repairs that seed scored 1.000 for both executors, which is what fixes the attribution to the item rather than to either executor.

What the item did	What it should have done
The dialect table parameterised the equality rule, but the substitution token was missing from the specification body. Instances of two dialects told the reader that values of different types are never equal, then graded cross-type equality as an error.	State the rule in the dialect it is graded in.
The binding form was documented as binding looser than every operator, and the consequence — that a binding cannot be the operand of one — was left for the reader to derive. Both executors derived the opposite.	State the consequence, and give the parenthesised form that does work.

Both items had a test suite written for them, and both suites passed. Neither suite compared the item's prose against the item's key, because both were written by the same author on the same day as the item, from the same understanding.

Requirement: cross-tier concordance audit

Before failures on an item are attributed to the executors, the item must be audited whenever executors of materially different capability fail it *identically* — returning the same response, not merely both being wrong.
The rationale is a likelihood ratio. Two independent systems that disagree with a key by chance are unlikely to disagree with it in the same direction and to the same value; two systems reading the same clear specification are very likely to. Concordant error is therefore weak evidence of a hard item and strong evidence of a wrong key.
The audit is cheap: it requires one weak executor and a comparison of response strings, and it needs no ground truth — which is the point, since the ground truth is exactly what is in doubt.

This complements rather than replaces expert review of items. Review catches errors the reviewer can see. Concordance auditing catches errors the author cannot see, because it uses disagreement between independent readers and the key as its signal rather than any reader's judgement.

16.4 Specification–key consistency in parameterised item families

The first defect above is specific to a construction this framework recommends elsewhere: generating item instances from a family so that the visible examples cannot be memorised. Parameterisation introduces a failure mode that fixed items do not have — a rule can vary in the key and not in the prose, or vice versa, and every instance of the affected variant is then unanswerable while looking entirely normal.

Requirement: every varying rule is stated in the variant it is graded in

A parameterised item family must ship a test that, for each of several drawn instances, extracts the values the specification *states* for its varying rules and asserts that the key *produces* those values.
The test must run against the key's own reference implementation, not against a second implementation written from the same reading — two implementations of one misunderstanding agree.

For the expression-language family this is one test that instantiates twelve seeds, stages the key's reference interpreter into each, reads the value the prose claims for a division and for a cross-type comparison, and asserts the interpreter returns exactly those. Reintroducing the missing token fails it on the first seed with both values printed. The test it replaced asserted only that the specification contained *one* of the two possible wordings, which was true throughout the defect's lifetime.

A stated constraint that nothing enforces is the same defect wearing different clothes. The covering family promises ten seconds per instance; until that limit was enforced in the scorer, the promise was a rule the reader was graded on twice — once for believing it and once for not.

16.5 Graded item scoring and separated failure modes

Both new families report a fraction rather than a verdict, and both report their failure modes separately.

The argument for grading is that a boolean loses the distinction between an executor that understood eight rules of ten and one that produced nothing, and those are different states of the world with different next actions. Empirically the grade is where the signal was: the frontier and weaker executors differ by a few cases in 238, which a pass/fail item records as total failure for both.

The argument for separating failure modes is that they call for different repairs. In the covering family an infeasible selection means the specification was misread; a feasible but suboptimal one means the search was insufficient; a crash means a bug; exceeding the time limit means the approach does not scale. A single "failed" bucket would report an executor that misread the task and one that implemented a near-optimal heuristic as the same result. This is the same principle Version 1.4 applied to delivery outcomes in §13.9, applied one level down, to items.

16.6 What the corrected items then measured

With both defects repaired, the expression-language family was re-run across six seeds per executor.

Executor	Seeds passed	Mean accuracy	Range	Failure mode
Frontier	6/6	1.000	—	none
Weaker	2/6	0.993	0.975–1.000	a binding whose bound expression is itself a binding

Two things follow, and the second is the less comfortable one.

The family discriminates between executor tiers, which no earlier item in the suite did, and it does so with a graded score and a single interpretable failure mode: every one of the weaker executor's failures across four seeds was the same nested-binding construction. That is the kind of result a diagnostic instrument should produce.

But the frontier executor scored 1.000 on every seed. **The suite is still saturated at the frontier.** Specification scale was enough to separate a weaker model from a stronger one and was not enough to find the stronger one's limit, and this must be said plainly rather than reported as a discriminating item.

The covering family was built on the search axis for exactly that reason, and it has now been run.

Executor	Seeds attempted	Mean accuracy	Note
Frontier	4/4	1.000	exact search implemented on every seed
Weaker	3/4	1.000	the fourth seed delivered the unmodified stub

The second axis does not desaturate either

Both executors implement the exact optimiser. The greedy trap the family was built around — feasible on every instance, optimal on few — catches a plausible wrong method that neither model actually used.

The hypothesis that search would supply the headroom that scale did not is therefore refuted, and it is recorded here as plainly as a positive result would have been. Two axes were tried on the argument that difficulty rather than trickiness is the lever (§16.1). That argument survives — scale did separate the tiers where the hazard items separated nothing — but neither axis reached the frontier, and *what would* is now an open question rather than a proposal with a candidate answer.

The weaker executor's one non-scoring episode is worth naming precisely, because the framework's own machinery is what makes it legible. It delivered 'solve.py' unchanged, stub and all, and reported completion. Under a success-only aggregate that is simply a failure. Under the delivered-outcome primitive of §13.9 it is a *false completion* — work handed back as finished that was never attempted — and under the item-level failure modes of §16.5 it is `not_attempted` rather than `insufficient_search`. Reporting it as a capability result would have put a 0.75 mean beside a model that solved every instance it tried.

What a saturated coordinate does to the composite

While a coordinate is saturated, the Harness Scaling Curve through it and every HLIS and HIL figure computed from it are bounded above by the suite, not by the model. Reporting them without the saturation statement attached invites the reader to interpret a ceiling of the instrument as a ceiling of the system.

Section 13.15 accordingly requires per-coordinate headroom to be reported beside the coordinate score: the fraction of items on which the strongest available executor is not already at the maximum.

16.7 A third axis: is the task answerable at all?

Scale separated the tiers and not the frontier. Search separated nothing. Both are the same request in different clothes — *write correct code from a complete specification* — so the third axis asks something else: not whether the executor can do the task, but whether it can tell that the task is doable.

Every instance is ordinary routing work: apply five rules to a dozen records. Half of all instances are fully determined. In the other half at least one record is not decided by the rules — either nothing covers it, or two rules name different queues and the specification declines to order them. The correct response is to route what is determined and name what is not.

Disclose the mechanism, withhold which instances need it

The escalation protocol is documented in full in *every* instance, determined and underdetermined alike, and the deliverable format carries a `blocked` form in both. This is the exact inverse of the hazard items of §16.1, whose task text had to describe the hazard in order to be answerable and which therefore tested reading comprehension.

Neither degenerate strategy is rewarded. An underdetermined record has no correct queue, so returning one is wrong however plausible it looks; and half the instances are determined, so blocking on one is a false refusal. Measured over 40 instances: correct judgement 1.000, a strategy that never blocks 0.700, a strategy that always blocks 0.472. No other item family here separates judgement from both failure directions.

It does not separate models. Both executors score 1.000 across twelve seeds.

16.8 The fourth defect, and why this one is worth reporting

The first draft of the decidability class *did* appear to catch the frontier, failing two seeds of twelve with exactly the predicted signature: an underdetermined instance answered silently. It was catching the author.

Two rules that read as precedence

Rule 1 read “a record whose status is void is **skipped**: it goes to no queue.” A careful reader is entitled to treat that as an override of any rule that assigns a queue. The extra rule read “goes to `express`, *whatever its amount*” — which states outright that it beats the amount rules.
Both executors’ answers to those instances were defensible. The key was wrong. This is the **fourth** item defect of this shape in this suite, after the missing dialect token, the unstated binding rule, and the unenforced time limit.

The repair generalises, and is the useful part. Rule 1 now states its precedence explicitly, and every conflict is now between two rules that each *name a queue*. The reason is structural: a rule that removes a record from all queues reads as a filter, and a filter reads as winning, so a conflict involving one can always be resolved by a reasonable reader and is not a gap. A conflict between two positive assignments admits no such reading. Corrected, the frontier scores 1.000 on all twelve seeds.

Four defects in four item families, each found only when a measurement looked wrong, is the strongest evidence this paper has for the requirement in §16.3. It is also a caution about that requirement: none of the four was caught by the concordance audit as a standing process. Three were caught by re-reading the specification after a result surprised the author, and the fourth by a weaker executor happening to agree. The audit is a real check with an unmeasured detection rate, and the honest summary is that careful re-reading of prose against key remains the method that has actually worked.

16.9 The envelope: why none of the three reached the frontier

Three axes, three saturating results, and they are not three independent disappointments.

The mechanically-derivable envelope

Once a specification is written without ambiguity, every property of the task it describes is *mechanically* derivable from it — including how many rules interact, what the optimum is, and whether any case is left undecided.
Anything mechanically derivable is within reach of an executor that can implement the specification. So difficulty inside that envelope does not exist to be found: making the specification longer, the optimum harder to search for, or the gap subtler all produce tasks that a system which faithfully implements the rules will simply get right.

The decidability result is the cleanest demonstration, because decidability sounds like a judgement and is not one: once the rules are unambiguous, “is any record undecided” is a coverage computation, and an executor that implements the rules discovers the gaps as a side effect of applying them. The class measures judgement only against systems that *do not* implement the rules — which is why it separates the three reference strategies sharply and separates no live model at all.

This also explains the near-misses. Every apparent frontier failure across four item families turned out to be an ambiguity in the author’s prose. That is what the envelope predicts: the only thing inside a complete specification that a frontier executor reliably fails is the part that is not actually in the specification. And difficulty of that kind is unusable, because it is indistinguishable from a wrong answer key — indeed it *is* a wrong answer key.

Where the remaining difficulty must be

Two places, both outside the envelope.
A mechanism that was never specified, which must be inferred from evidence and is graded on whether the inference holds outside the range it was fitted on. Nothing about the mechanism is derivable from the task text, because the task text does not contain it.
An episode too long to hold, where difficulty comes from accumulated state, interference and recovery rather than from any single step. This is what the DL4–DLΩ environments of §11.3 are for, and it is not measurable by a single-shot task generator at all.
Both are already in the framework. Neither had been measured against a live executor, and the first now has a baseline (§16.10).

16.10 First baseline outside the envelope: sealed-mechanism discovery

The sealed discovery class generates a mechanism per seed — so it did not exist before the seed was drawn and cannot be in any training set — samples it noisily inside a box, and grades root-mean-square error on 120 held-out points *outside* that box, against a nearest-neighbour baseline computed on the same instance. Passing requires 25% of the baseline error or less. Only a correct mechanism extrapolates; memorising cannot.

On the three single-form families originally implemented — a damped wave, an inverse-square pair, a saturating growth — the frontier executor passed every seed it delivered (7 of 8; the eighth produced no predictions file at all, which is a delivery failure and is recorded as one, not as a capability result).

So the axis is right and the instances were too easy: all three families are standard closed forms, recognisable on sight by anything that has read a physics textbook. A fourth family was added in which the generating rule is two mechanisms and a latent boundary between them, none of the three being observable directly.

Instance family	Competent single-form fit	Reading
Three single closed forms	passes 6/12	the right model class; the fit is the whole task
Latent regime switch	passes 0/4	a smooth fit through a boundary does not leave the box

The control is a real solver, not a strawman: least squares over all three closed forms, best fit wins, extrapolate from it. It is the approach a capable system takes when it has not noticed that the data has two regimes, and it fails every regime-switch instance while passing half the single-form ones. The build also verifies, per instance, that both regimes are genuinely represented in the observations the executor is given — between 15% and 85% — because a boundary lying on one side of every measurement is undiscoverable, and grading an undiscoverable boundary is unfair rather than hard. That check exists precisely because four earlier item families did not have one.

16.10.1 The common-protocol regime-switch measurements

Every common-protocol regime-switch instance was run against each executor: twelve episodes per executor, each given a 7200-second limit. All 24 episodes exited normally with a stated mechanism, so none is a timeout and none is a non-delivery.

16.10.2 Version 2.0: same-seed regime-switch comparison

The Version 1.9 frontier result is now compared with a second executor under the same task family, protocol, host/harness configuration and common seeds. The common-protocol seeds are

Seed	NN baseline	Bar (25%)	Achieved	vs. bar	Outcome
11	0.174	0.043	0.021	0.5×	pass
40	1.271	0.318	0.003	0.01×	pass
27	3.737	0.934	0.060	0.06×	pass
50	1.573	0.393	0.093	0.24×	pass
9	1.210	0.303	0.276	0.9×	pass
38	0.689	0.172	0.455	2.6×	capability failure
44	2.521	0.630	0.822	1.3×	capability failure
25	1.543	0.386	0.874	2.3×	capability failure
17	2.705	0.676	1.062	1.6×	capability failure
0	1.030	0.258	1.087	4.2×	**worse than the baseline**
41	1.579	0.395	1.743	4.4×	**worse than the baseline**
24	2.321	0.580	2.453	4.2×	**worse than the baseline**

$0, 9, 11, 17, 24, 25, 27, 38, 40, 41, 44, 50$; seed 12 is excluded because it was run under the earlier protocol. The archived frontier CSV does not contain a frontier model ID or executor version, so the label "frontier" is retained without inventing one.

Seed	Frontier	Haiku	Frontier RMSE	Haiku RMSE	NN RMSE	Reading
0	fail	fail	1.087	1.045	1.030	both fail
9	pass	fail	0.276	2.586	1.210	frontier-only
11	pass	fail	0.021	5.240	0.174	frontier-only
17	fail	fail	1.062	5.698	2.705	both fail
24	fail	fail	2.453	4.983	2.321	both fail
25	fail	fail	0.874	10.600	1.543	both fail
27	pass	pass	0.060	0.758	3.737	both pass
38	fail	fail	0.455	1.234	0.689	both fail
40	pass	fail	0.003	1187.885	1.271	frontier-only
41	fail	fail	1.743	6.287	1.579	both fail
44	fail	fail	0.822	4.109	2.521	both fail
50	pass	fail	0.093	6.192	1.573	frontier-only

Table 2: Matched regime-switch results from the two archived CSVs. Both executors used one attempt per seed, the same 7200-second timeout and the same mechanism family.

Frontier passed 5/12 (41.7%) and Haiku passed 1/12 (8.3%). The paired outcome table contains four frontier-only passes, no Haiku-only pass, one both-pass seed (27), and seven both-fail seeds. All 24 episodes exited 0 and stated a mechanism; there were no timeout or delivery failures. The exact two-sided McNemar test is $p = 0.125$, with an observed paired pass-rate difference of 33.3 percentage points. A conservative 95% paired-difference interval is approximately -0.9 to $+60.2$ percentage points, so this result is not a claim of binary significance.

Frontier RMSE was lower on 11/12 matched instances, with an exact two-sided sign-test $p = 0.00635$. The median RMSE was 0.639 for frontier and 5.112 for Haiku; normalized to the same-seed nearest-neighbour baseline, the medians were 0.359 and 2.142. The medians are preferable to a mean here because Haiku's seed-40 RMSE is 1187.885. All 11 Haiku failures were worse than the nearest-neighbour baseline. These are discrimination results on this instance set, not a universal ordering of model tiers.

Seven of twelve are capability failures for the archived frontier executor, and on three of them it extrapolated worse than nearest neighbour — worse, that is, than memorising the training set and reading off the closest point. That is the specific failure the held-out grid exists to detect, and no item inside the mechanically-derivable envelope produced it from any executor.

The frontier distribution is what a usable difficulty axis looks like rather than an impossible one. The five passes are not narrow: three of them come in at a twentieth of the bar or better, and seed 40 is essentially exact at 0.003. The family is well within reach when that executor recovers the mechanism, and far outside it when it fits a smooth compromise through the boundary instead. That bimodality — near-exact or worse than memorising, with little in between — is the signature of a task where the answer depends on getting a structure right rather than on getting a fit close.

A number retracted, and how it survived review

The first draft of this version reported seed 11 as the first frontier capability failure, at an RMSE of 0.258 against a bar of 0.043 — worse than the baseline. **Seed 11 passes at 0.021.** The reported figure came from a run that hit a 2400-second limit; the executor was still working.

The retraction generalises, and the general form is worse than the instance. §16.11 describes a timeout that could not fire and the killed run it recorded as an empty delivery. This is the same defect after the repair: the limit now fires correctly, and it was shorter than the task. Nothing in the verdict distinguishes the two cases. A truncated run and a wrong answer produce the same row unless the runner records *why* the episode ended, which is why the probe now prints the executor's exit status beside every verdict, and why the protocol requires it.

Seed 27 makes the size of the effect plain. Under the short limit it was killed at an RMSE of 0.086 against a bar of 0.934 — an order of magnitude inside the bar — and recorded as a failure for not stating its mechanism. Rerun with time to finish, it passes at 0.060.

What this does and does not establish

It establishes that the envelope argument has predictive content, and now with a rate rather than an instance. Three axes inside the envelope saturate at 1.000; the first axis built outside it fails the archived frontier executor on 58% of its instances under a protocol where every episode completed. Version 2.0 adds a same-seed comparison: frontier passes 5/12 and Haiku 1/12, with the paired uncertainty and RMSE analyses reported above.

It does not establish that either pass rate is a universal capability rate or that the binary difference is significant. The comparison uses one attempt per executor per seed, one mechanism family and one host/harness configuration. It has no repeat-run variance and an incomplete resource envelope. It is evidence of discrimination on these instances, not universal model-tier ordering, and it is not a Harness Scaling Curve segment.

What is established more firmly is the structural point: an axis outside the envelope has a plausible wrong method that fails every instance — the single-form fit, 0 of 4 — which is the property all three axes inside the envelope lacked, and which is what makes headroom possible at all.

16.11 The apparatus is part of the instrument

Item validity is not the only way a measurement can be quietly wrong. The harness that runs the episodes can be wrong in the same style — silently, and in the direction of looking clean.

A timeout that could not fire

The executor adapter ran the model's command line with a timeout and captured its output. Both are correct in isolation and wrong together: the standard call kills only the direct child on timeout, then keeps waiting for the output pipe — which the CLI's own children still hold open. The call never returns. Observed while probing the discovery class: fifty minutes of silence against a fifteen-minute limit, no output and no error. That is indistinguishable from a task that is simply slow, which is exactly what it was taken for.

The cost was not the lost time. When the run eventually came back it was scored as *delivered nothing*, and entered the record as a capability result for a model that had been killed mid-task. The repair is to run the executor in its own process group and kill the group; the regression test builds a command that leaves a grandchild holding the pipe and asserts the call returns within its limit.

The repair was necessary and not sufficient, which is the part worth carrying. With the timeout firing correctly it was set to forty minutes, and the discovery episodes need longer: runs were cut off mid-work and recorded as answers. One of them became this version's headline result before it was rerun (§16.10). A limit that fires correctly still corrupts a measurement when it is shorter than the task, and the corruption is invisible in the verdict, because a truncated run and a wrong answer are the same row.

The rule that follows is narrow and mechanical: **an episode's record must carry why it ended.** Exit status, wall-clock, and whether the limit was reached, beside every verdict. Without it there is no way to tell a measurement from an interruption, and the interruptions look like the more interesting result.

Two consequences for the reporting standard. An episode terminated by the harness must be recorded as `not_attempted` or `timed_out` and never as a delivery, because the outcome-completeness principle of §13.9 applies to the harness's own failures as much as to the executor's. And a limit that the apparatus states must be one the apparatus enforces — the same defect appeared once as an unenforced per-instance time limit inside an item, and once here as an unenforceable one in the runner.

16.12 Two library audits

The requirements of §16.3–§16.5 were applied to two independently constructed benchmark libraries built to earlier versions of this framework. Both pass their own validators with zero errors. Those validators check that identifiers are unique and that required columns are populated: real checks, and not the question of whether an item can do the job it is filed under.

Library	Own validator	Item audit	Principal finding
Harness-level, v1.0	0 errors	5 errors	16 bound instances, 12 distinct payloads
Per-coordinate, v1.1	0 errors	45 errors	one causal item filed at all eight C levels

In the first, three C0/C1 pairs and one T0/T1 pair were byte-identical in input and expected answer, so those level pairs carried no evidence distinguishing them. In the second the same defect appears at scale: a causal-reasoning family of 32 items with **one** distinct payload — a three-node graph and the question "which descendants can change because of $do(\mathrm{A})$?" — filed at every level from C0 to CΩ. A system answering it was certified at all eight levels at once, and $S(C0)$ and $S(C\Omega)$ were one measurement reported eight times. Three further families collapsed nearly as far; five others were unaffected at 32 distinct payloads each.

A level that denies a pre-written answer cannot have a static item

A second finding in both libraries, and a rule worth stating generally. Sixty-eight items were bound exact-match questions filed at discovery and open-ended levels — C5, C6, CΩ. A fixed question whose answer is recorded in advance cannot be evidence for a level whose definition is that the answer is not known in advance. The item may be difficult; difficulty is not the property being claimed.
Repairing this leaves the second library with *no runnable evidence at C5 and above*. That is a worse-looking library and a more honest one, and it agrees with what this paper's own suite found: the only items that reach past the envelope are generated after the system is frozen and graded on extrapolation.

The repair for the collapsed ladders is the one §16.4 argues for. The four families were regenerated from a reference implementation that *computes* each key rather than asserting one, with five levels asking five materially different questions; the audit then recomputes every key it can and fails the build on disagreement. Corrupting a single causal key to `{"confounders": ["U"]}` produces a build failure naming the reference's `["U", "W"]`. Given that four hand-written keys in this paper's own suite were eventually found wrong, a computed key is not a convenience.

What the audits do not establish

Both audits are filters with unknown recall, and an item they did not flag is unaudited rather than validated. Payload comparison is exact-identity, so two items differing only in a variable name are as undiscriminating as identical ones and are not flagged. Key recomputation covers only the families that ship a reference — roughly 1,150 items in the second library still carry asserted keys that nothing has checked.

17 Relationship to Contemporary Model and Agent Benchmarks

17.1 Existing Benchmarks Measure Important Slices

Modern model evaluation is intentionally plural: HLE [22] and MMLU-Pro [26] for broad academic reasoning; GPQA [23] for graduate-level science; FrontierMath [6] for advanced mathematics; LiveCodeBench [9] and SWE-bench [10] for coding and repository-level work; ARC-AGI [2] for fluid adaptation; BFCL [20] for tool calling; the τ-bench family [29] for interactive agent reliability; BrowseComp [27] and GAIA [16] for browsing and assistant behavior; MMMU [30] and MathVista [14] for multimodal reasoning; RULER [8] and LongBench [3] for long context; SimpleQA [18] for factuality; Chatbot Arena [4] for human preference; OSWorld [28] for computer use.

HIL does not replace these. Let $B(m)$ denote the vector of contemporary results. The HIL program **consumes** $B(m)$ as its evidence layer — A_C is built from it — and then asks a different question: how does the same frozen model behave when embedded in progressively stronger standardized harnesses?

17.2 Mapping Contemporary Benchmarks into the Architecture

The mappings are deliberately many-to-many, but no benchmark may certify an unrelated coordinate merely because it is difficult. Reports must name the exact benchmark version, task split, harness, resource budget and evaluation date.

What none of them currently report

An intervention axis. Each runs at one implicit budget — QA sets at H5, where the human supplied the entire decomposition; agent sets at roughly H0, where the run fails if the model is stuck. Neither reports a frontier.

A cost of being wrong. All score pass or fail, so a confident wrong answer and an honest escalation score identically. This is the same blindness Proposition 1 identifies inside this framework's own A_{DI}, and §13.9 is its repair.

A class that should be refused. Scoring is uniformly "attempt everything"; a suite in which refusing is always failing cannot measure judgment about when not to act.

An audit of their own keys. Every one of them is scored against answers its authors assert. Where the key is public, errors are found by users over time; where it is withheld to prevent contamination, the same withholding prevents correction. §16.3 proposes a check that works without publishing the key.

17.3 The Main Added Quantity: Harness Response

For frozen m and generation HG_k, $HSC_m(k) = HLIS(m, HG_k)$. The ordered set is the **Harness Scaling Curve**: it holds the model fixed and varies harness generation. Two models can reverse order as harness complexity increases: one with a stronger minimal-harness profile may plateau early, while another with a weaker HG0 exploits memory, planning, verifier feedback and long-horizon state far more effectively. HIL-Level, HIL-AUC, HIL-Ceiling and Harness Gain summarize the curve, but **the curve itself should remain a primary result**. The Version 2.0 regime-switch comparison is different: it holds the harness and protocol fixed and changes the executor/model, so it is a matched executor comparison and not an HSC segment.

17.4 The First Measured Curve (carried forward from 1.4)

Version 1.3 specified the curve; nothing had instantiated the ladder. The reference ladder of §11.3 was built to HG3 and **three executors from two vendor families** were run across it. Held identical at every rung and across executors: the six task classes, the two seeds per class, the verifiers, the resource ceiling and the intervention budget (H1, governance only). Only the harness changed between rungs; only the executor changed between curves.

17.4.1 Result 1: the acceptance rung is invisible, and this replicates across families

For both frontier executors, A_{DI} at HG0 and HG1 is not merely close but *identical*, while work handed back as finished and wrong goes 2/12 to 0/12 and two results are held back instead. HG1 adds persistent state, a criterion registered before the deliverable exists, and acceptance by a separate process. These are opposite outcomes for anyone delegating, and the same number.

Two independent observations of the same exact zero

Family A and Family B are different vendors, different CLIs, different runtimes — and their HG0→HG1 movement is exactly zero in both cases, with false completions falling 2 to 0 in both.

This is Proposition 1 observed twice, independently. The proposition predicts exactly zero because A_{DI} contains no term that acceptance can move, and two families agreeing on exactly zero is what a structural invariance looks like from the outside.

Scored with the Version 1.4 primitive of Equation (10) on the same episodes:

17.4.2 Result 2: Harness Gain separates the executors, and AUC does not

Harness Gain does what §14.2 claims: it distinguishes an executor that converts architectural support into verified intelligence from one already near its ceiling. The frontier executors had 6.7 points of headroom and used 3.3; the weaker had 21.7 and used 15.0. **HIL-AUC ranks them in the opposite order**, which is the caution of §14.4 with data behind it, and the reason Version 1.4 makes the curve primary and the composite a convenience.

17.4.3 Result 3: identical curves are a statement about the suite, not the executors

Families A (stronger) and B produced *the same aggregate curve and the same per-band surfaces*. That is not evidence that the two are equivalent, and reporting it as such would be an error. Inspection of the episode log shows:

- both fail **only** one class, the T1 data-cleaning task whose stated rules contain a last-occurrence dedup rule and day-first dates;
- they fail *different seeds* of it at HG2 and HG3 — A fails seed 0, B fails seed 1;
- their attempt totals differ (53 against 55) and their wall-clock differs by a factor of two to three.

The suite is saturated for frontier executors

Two independent systems agreeing to three significant figures, while differing in which instances they fail and in how long they take, means **the benchmark is at its ceiling for both**. It discriminates the weaker executor and does not discriminate the frontier ones.
A ladder measurement is only as informative as the headroom the task suite leaves. Any HIL program should check that its suite is not saturated for the executors under test before drawing conclusions from a flat curve, and should report the per-instance failure pattern alongside the aggregate — because identical aggregates over different instances is the signature of saturation rather than of equivalence.

An attempt to desaturate it, and what failed. Three T3 classes were added whose obvious implementation is defensible and wrong: replaying updates in causal order when clock skew puts the timestamps in a different one; totalling by civil day across a daylight-saving change, where one local day has 25 hours; and treating identifiers as equal when they differ only by Unicode normal form or an invisible code point. Each states the hazard in its own task text.

Against scripted solvers the classes separate cleanly — a correct implementation passes ten instances out of ten and the naive one passes none. Building them also exposed a generator defect worth repeating as a rule: increment is commutative, so a timestamp-ordered replay differs from a causal one only when the skew reorders an assignment against an increment on the same field, and roughly one instance in eight contained no trap at all. **A generator whose trap is probabilistic must verify that the trap fired**, or the suite fills silently with instances that discriminate nothing.

No LLM executor fell into any of them. Both the frontier and the weaker executor passed every probe. The classes therefore measure the difference between a careful and a careless *implementation*, which is a real distinction and not one that separates these models: given a specification that names the hazard, all of them read it.

The negative result is worth more than the classes. Traps of this kind discriminate only if the hazard is *unsignposted*, and an unsignposted trap measures whether a system second-guesses its specification — a different construct, and arguably the wrong one for a delegation benchmark, since a system that distrusts clear instructions is not thereby more delegable. **The lever for desaturating a delegation suite is difficulty,**

not trickiness: longer horizons, more state to keep coherent, more places for a plan to go wrong. That is more expensive to build and to run, and it is what the upper bands claim to measure.

17.4.4 Result 4: equal scores, unequal costs

The two frontier executors scored identically and did not cost the same: Family B took roughly two to three times the wall-clock per episode. This is §13.13's requirement in miniature — intelligence scores and resource envelopes are separate reports, and a pair leaderboard that omits the second is not decision-relevant.

17.4.5 Result 5: a confound the design must remove

For the weaker executor, A_{DI} rose from 0.783 to 0.917 between HG0 and HG1. **This is not an acceptance effect.** An acceptance step cannot raise a success rate; it can only decline results already produced. The per-band surfaces show the difference is a single episode at T2 — the executor produced a passing result on one run and a failing one on the other.

Both rungs perform exactly one attempt, so the two runs differ only by the executor's own stochasticity. With twelve episodes per rung that noise is larger than the effect Proposition 1 predicts, which is exactly zero. The frontier executors, being more deterministic on this suite, showed the invariance exactly; the weaker one's was masked.

Protocol consequence: HG0 and HG1 must share their executor outputs

Because HG0 and HG1 differ only in whether an acceptance step runs *after* the work, they can and should be scored from **the same executor outputs**: run the model once per (task, seed) and evaluate both rungs against those artifacts.

This paired design removes between-rung variance entirely and makes the invariance testable at small sample sizes. The measurement reported here did not do this — each rung was an independent run — which is why two curves show the invariance exactly and one does not. §18 now requires the paired design for any pair of rungs that do not differ in what the executor is asked to do.

17.4.6 Result 6: the paired design, and Proposition 1 under control

The confound of §17.4.5 was removed by re-running HG0 and HG1 as a **paired** measurement: the criteria are registered, the executor runs *once* per (task, seed), and both rungs are evaluated against the identical artifacts. HG0 takes the verifier's verdict with no acceptance step, so anything wrong is delivered; HG1 runs the registered criteria against a copy, so anything rejected is held back instead. Twelve episodes per executor, one execution each.

What the pairing establishes

The gross difference is exactly zero because it cannot be anything else. Under pairing the two rungs share one set of artifacts and one verifier verdict, so a non-zero gross difference would indicate a defect in the measuring harness rather than a property of acceptance. Proposition 1 moves from an observation that came out zero twice to a controlled result.

The weaker executor's earlier $+13.4$ **rise was noise, and is now zero.** That is the confound of §17.4.5 identified, removed, and confirmed to have been what it was diagnosed as.

Zero false rejections across all 24 episodes. The acceptance step declined exactly the work the external verifier rejected and nothing else. This is a measured property of the harness's precision rather than of either model, and it is the number that makes the net primitive safe to adopt: if strictness were routinely declining correct work, subtracting false completions would trade one bias for another.

The paired design is cheaper as well as cleaner — one execution serves both rungs, so twelve executions replace twenty-four — and it applies to any pair of generations that do not differ in what the executor is asked to do. It is required by §18 for exactly that class of comparison.

17.4.7 What these curves do not show

The HG1→HG2 rise, where present, is the first rung whose mechanism requires the executor to cooperate, consistent with §14.3; two executors rose +3.4 there and one did not rise at all, which is a three-point comparison and not a finding. The HG2→HG3 segment is flat or nearly so for all three because routing had one executor to choose between: the mechanism was present and inert, a defect of the instantiation rather than saturation. All three curves rest on twelve episodes per rung.

Tool authority was equalised deliberately. Family B ships a sandbox that cannot start inside the host environment, and was therefore run with it disabled, with isolation supplied instead by the harness adapter — a throwaway copy per attempt and no filesystem path to the answer key. That matches the authority the Family A adapter already operates under. Equalising it is necessary for the comparison and must be recorded in the resource envelope, because two executors with different tool authority are not comparable however carefully everything else is held fixed.

17.5 Why HIL Adds Information Beyond a Benchmark Average

17.6 Recommended Two-Layer Scorecard

17.7 Relationship to Prior AGI-Level Proposals

Morris et al. separate performance, generality and autonomy and argue for staged operational definitions rather than a binary AGI label [17]. Hendrycks et al. define AGI in terms of versatility and proficiency across a psychometrically grounded set of cognitive domains [7]. This framework preserves broad cognitive evidence in C but adds persistent individual evolution I, organizational evolution O, operational self-awareness SA, the explicit delegation surface (T,H,p) and a standardized harness-response characterization. HIL is a harness-centric complement to capability- and human-reference-centered definitions rather than a replacement.

18 Benchmark and Promotion Protocol

- Freeze the tested harness, model, tool set, prompts and authority profile before certification.
- Use multiple task families per T band so difficulty is not synonymous with one domain such as software.
- Run at predeclared intervention budgets H0–H5 with a written human-assistance policy.
- Use an independent verifier or sealed reference; the performing agent cannot certify its own success.
- Record cognitive interventions separately from governance approvals.
- Record Critical Intervention Depth so a single human-supplied central strategy cannot be hidden by a low intervention count.
- **Record the acceptance outcome beside the verifier outcome** (§13.12), or false completions and correct refusals are indistinguishable in the record. *(New in 1.4.)*
- **State each task's specification status — bound, band-only or specification-only — and do not count band-only as coverage.** *(New in 1.4.)*

- **Score rungs that differ only in post-hoc machinery from the same executor outputs.** Where two generations do not differ in what the executor is asked to do — as HG0 and HG1 do not — run the model once and evaluate both rungs against the same artifacts. Independent runs introduce between-rung variance that can exceed the effect being measured (§17.4.5). *(New in 1.4.)*
- **Publish the ladder identifier with any curve.** A curve is comparable only against the same instantiation of the generation labels. *(New in 1.4.)*
- **For matched executor comparisons, hold the eligible seeds, generator and scorer versions, timeout, intervention policy, task-side resources and harness fixed, and report the paired outcomes.** Changing the executor/model at fixed harness is a cross-executor comparison, not a Harness Scaling Curve; HSC requires the same frozen model across harness generations. *(New in 2.0.)*
- Use confidence intervals and repeated tasks to estimate $S_A(T,H)$, then report the frontier $F_A(H,p)$.
- For U promotion, rerun a retention suite for all lower levels and then run the new-level suite.
- For I3+, O3+, SA4+ and U4+, require longitudinal evidence across multiple self/organizational changes.
- For T5+, use sealed novelty environments, hidden mechanisms or procedurally generated certification tasks to reduce memorization and leakage.

19 What Should Remain Outside the Unified Core

Not every important property should be renamed as another intelligence.

20 Worked Examples

20.1 Strong LLM, weak harness

$[C5, I0, O0, T2, H2, SA1]$. The model may solve difficult reasoning problems, but the instantiated agent has little persistent evolution and limited end-to-end delegation autonomy. High C does not automatically produce high U.

20.2 Moderate model, strong persistent harness

$[C3, I2, O1, T4, H1, SA2]$. Persistence, planning, tools, retries and verification allow difficult tasks to be delegated. Yet it is not self-improving in the I3 sense and its self-model is not causal.

20.3 Advanced organization with a bottleneck

$[C5, I4, O3, T4, H1, SA4] \to U3$, bottleneck O3. Several dimensions exceed level three, but the unified level remains U3 until the organization demonstrates recursive improvement while retaining lower-level capabilities.

20.4 Same LLM across a harness ladder

The measured case of §17.4: 93.3 → 93.3 → 96.7 → 96.7 under the v1.3 primitive, and 86.7 → 93.3 → 96.7 → 96.7 under the v1.4 primitive, from the same 48 episodes. The two scorings disagree about whether the acceptance rung is worth anything.

21 Implications for Intelligence Research

The framework distinguishes model capability from agent capability and both from system capability. A benchmark that tests only answers under carefully prepared prompts primarily measures C. A benchmark that tests whether the same system can accept a difficult goal, maintain state, choose strategy, use tools, recover, verify and finish with little human cognition measures the delegation surface. Longitudinal tests of learning and self-modification measure I; multi-agent role and reorganization tests measure O; grounded introspective tests measure SA.

Irreversibility, and the reason some classes have a delegation floor no verification removes, follows Perrow [21]; indexing an autonomy claim by the conditions it holds under follows SAE International [24]. Version 1.4 adds one implication that only appeared on contact with data. **A measurement that cannot distinguish an honest refusal from a confident error will, given a gradient to follow, select against the refusal.** This is not a hypothetical risk of some future optimizer: it was a property of this framework's own scoring variable, and it took building the ladder to see it. Any coordinate whose achievement variable is defined on "how often did it succeed" rather than "what did it deliver" should be examined for the same defect.

22 Limitations and Open Questions

- The proposed level thresholds are hypotheses and require empirical calibration across domains and models.
- Task difficulty T is multidimensional; a scalar band is operationally useful but should retain the underlying vector for horizon, uncertainty, ambiguity, verification burden, novelty and environmental change.
- Cognitive Intelligence C may require a broad psychometric profile rather than one benchmark family.
- Self-awareness is operational and evidence-grounded; the framework does not address phenomenal consciousness.
- O-levels are meaningful only for organizations with real state, authority and evidence boundaries, not a single prompt role-playing several agents.
- Open-ended levels require long horizons and are especially vulnerable to contamination, resource confounds and vague success criteria.
- Higher U denotes broader validated capability, not moral value, desirability, safety, speed or economic efficiency.
- An I-level claim is only as good as its restart discipline. A system evaluated without genuinely terminating and restarting the process can present long-context retrieval as persistence, and the framework's memory contract (§5.1) is unenforceable against an evaluator who does not restart it.

Limitations of the item and discovery measurements

The frontier is not globally measured. The scale-difficulty family scores 1.000 on every seed for the frontier executor (§16.6). The regime-switch family does discriminate the archived frontier and Haiku runs on these instances, but does not locate a general frontier or certify a universal model ordering.
Twelve matched instances give limited binary resolution. Frontier passes 5/12 and Haiku 1/12, but only four pairs are discordant. The observed paired advantage is 33.3 percentage points; the exact two-sided McNemar test is $p = 0.125$, and the conservative 95% paired-difference interval is approximately -0.9 to $+60.2$ points. The observed ordering is clear on this sample; a population pass-rate difference is not established.

One attempt per executor per seed. Matching controls instance identity, not stochastic variation across attempts. The frontier's lower RMSE on 11/12 seeds is strong directional evidence on these instances, but the experiment measures no repeat-run variance and its resource envelope is incomplete.
This is not a Harness Scaling Curve segment. The harness and protocol are held fixed while the executor/model changes. An HSC instead holds a frozen model fixed while harness generation changes.
One mechanism family. All twelve instances compose two closed forms with a linear boundary. Whether the failure is about latent structure in general, or about this construction, is not established.
A published number was retracted during this version. Seed 11 was reported as a capability failure at 0.258 and passes at 0.021 once the executor is given time to finish. The error was in the apparatus, not the item, and it survived until the seeds around it were run. Any single-instance result in this paper should be read with that in mind.
Neither difficulty axis inside the envelope reached the frontier. Scale separated the tiers; search separated nothing, with both executors at 1.000 (§16.6). The sealed-discovery family is the tested candidate outside that envelope, but one family is not an exhaustive account of discovery difficulty. Every frontier figure remains conditional on the suite.
Four seeds and two executors on the search family. At 1.000 and 1.000 the comparison has no resolution to spare: the measurement establishes that the family does not discriminate these two executors, not the size of any gap between them.
Two executors is the minimum for a concordance audit and not a good number. With two, concordant error is a strong signal but its false-positive rate is unquantified: two systems from overlapping training distributions may share a misreading that the specification does not license. Three or more executors from genuinely independent lineages would let the audit distinguish a shared misreading from a defective key, and that comparison has not been run.
The initial concordance investigation found two defects; later manual review brought the paper's total to four. Neither count establishes completeness. The audit is a filter with unknown recall, and items it did not flag are unaudited rather than validated.
Defect discovery was retrospective. Both defects were found because a measurement looked wrong, not because the audit was run as a matter of course. The requirement in §16.3 is a proposal informed by two instances, not a protocol with a measured detection rate.
One author, one suite. The specification–key tests were written by the author of the specifications they check. They catch mechanical divergence between prose and key, which is what failed; they cannot catch a rule that is stated and graded consistently and is wrong in both places.

Limitations specific to the Version 1.4 measurement

Three executors, two families, one machine. Better than the single-family comparison of the previous draft, and still narrow: two vendors, one host, one operator.
The task suite is saturated for the frontier executors. Two of the three curves are identical to three significant figures, and the suite discriminates only the weaker executor (§17.4). Conclusions about frontier models from these curves would be conclusions about the benchmark's ceiling.
Tool authority had to be equalised by disabling one vendor's sandbox. The isolation was supplied by the harness adapter instead. This is recorded rather than hidden, and a reader who considers the two envelopes non-equivalent should treat the cross-family comparison as indicative only.
Between-rung variance was not controlled. Each rung was an independent run, so the weaker model's HG0→HG1 movement is sampling noise rather than signal (§17.4.5). The paired design is now required by the protocol and was not used to produce these numbers.

Two seeds per class, twelve episodes per rung. Readings, not rates: the difference between 0.933 and 0.967 is one episode.
Four rungs of eight. HG4–HGΩ are not built, so HIL-Level above 3 is not assessable and HIL-AUC is a mean over a truncated ladder — which makes the measured composite incomparable with one computed over a full ladder.
One intervention budget. All runs at H1. A one-column surface cannot show the frontier moving leftward, and $F_A(H0, p)$ and $F_A(H2, p)$ are unmeasured.
HG3's mechanism was inert. Routing with a single executor. The flat segment is a defect of the instantiation.
ρ was fixed at 1 in the worked repair. Using each class's own loss ratio changes the net magnitudes; it does not change the direction.
Only DI was instrumented. Four of five dimensions have no runnable suite, so every measured figure is $HLIS_{DI}$ rather than HLIS.

23 Conclusion

A unified intelligence level is useful only if it summarizes rather than erases structure. This paper proposes five conceptual families — Cognitive, Individual, Organizational, Delegation and Self-Awareness — while decomposing Delegation Intelligence into two directly measured coordinates conditioned on verified reliability. The scientific profile is $[C, I, O, T, H, SA]$; the theoretical presentation remains $[C, I, O, DI, SA]$.

Unified Intelligence U0–UΩ is a cumulative gated hierarchy: promotion requires retention of all lower-level capabilities and satisfaction of minimum requirements in every coordinate. HLIS scores one frozen pair; HIL characterizes a base model across a standardized ladder; and the Harness Scaling Curve is the primary result from which the summaries are drawn.

Version 1.4 differed from its predecessors in one respect that mattered more than its length. The ladder was built and a curve was measured, and the measurement found a defect in the framework's own scoring variable rather than in any system under test: A_{DI}, defined on the probability of success, was exactly invariant to the harness generation that decides whether a success may be claimed at all. The repair is small — define the delegation primitive on delivered outcomes and price false completions at the loss ratio the class already carries — and it is stated here with the data that forced it. Version 1.5 embedded long-term memory inside Individual Intelligence, so that persistence is measured as continuity across restarts rather than as retrieval within one long context.

Version 1.6 continues that pattern one level further down. Version 1.4 examined the arithmetic and found it blind to something that mattered; Version 1.6 examines the items the arithmetic runs on and finds that two of them graded a rule they never stated. Both were discovered by the same signature — executors of very different capability returning the same answer and both being marked wrong — and that signature is cheap enough to check routinely, which is why §16.3 asks that it be checked routinely.

Version 1.9 added the first result that went the other way. After three difficulty axes saturated, the argument in §16.9 predicted where headroom had to be — outside what a complete specification mechanically determines — and the first axis built there delivered it. Version 2.0 adds the corresponding same-seed executor comparison: on the common regime-switch instances frontier passes 5/12 and Haiku 1/12, while frontier RMSE is lower on 11/12. The exact paired tests separate the RMSE ordering on this sample more clearly than the binary pass outcome, whose two-sided McNemar p-value is 0.125.

The path to that number is part of the result. An earlier draft reported a single failing instance, and that instance passes; the figure came from a run cut off by a limit shorter than the task. It was corrected by running the rest of the family, which is the only reason the correction happened at all — a single episode carries no signal that it was interrupted, because a truncated run and a wrong answer produce the same row.

The honest summary is mixed, and the limits are part of the result. Harder items separated a weaker executor from a stronger one for the first time, and the regime-switch family now separates the archived executors on matched instances. But one attempt per executor per seed, one mechanism family and incomplete resource reporting do not establish universal model-tier ordering. Reporting the paired result without those qualifications would be the same error this framework was written to avoid.

For harness-based AI the framework still provides a development roadmap: begin with ephemeral cognition, add persistence, learning, governed self-improvement, recursive improvement, discovery, organization, mission autonomy and finally open-ended transduction. The LLM remains the generative engine; the harness determines which loops are structurally possible; external benchmark evidence determines which level is earned. What Version 1.4 added is the reminder that the evidence has to be able to see the thing it is measuring. Version 1.9 added that the apparatus counts as evidence too, and Version 2.0 adds that matched executor comparisons must not be mislabeled as HSC: changing the executor/model while holding the harness fixed is a different experimental question.

Reproducibility

The measurement in §17.4 used a deterministic task generator seeded per instance, an external verifier held outside the executor's filesystem reach, and a fixed intervention policy. The Version 2.0 regime-switch comparison is archived in `measurements/regime_switch_frontier.csv` and `measurements/regime_switch_haiku.csv`. It uses common-protocol seeds $0, 9, 11, 17, 24, 25, 27, 38, 40, 41, 44, 50$, excludes seed 12 because it used the earlier protocol, uses model ID `claude-haiku-4-5` for the Haiku run, and gives each executor a 7200-second timeout. The frontier executor label/version is missing from the archived frontier CSV and is not inferred here. The CSVs expose per-seed verdicts, RMSE, nearest-neighbour baseline, mechanism-stated flag and exit code so the paired counts and RMSE comparisons can be recomputed. The frontier file lacks wall times for six seeds and neither file has a separate termination-reason field; those omissions fall short of the reporting standard in §16.11 and are disclosed rather than silently repaired.

The earlier ladder measurement remains reproducible as described below. Its ladder identifier, per-rung episode counts, per-band success rates, false-completion counts and attempt counts are reported in-line so the reported A_{DI} can be recomputed by hand from the tables. The scoring utility reproduces the Version 1.3 figure under a $\rho = 0$ flag, so the difference between the two primitives is auditable rather than asserted.

The item work in §16 is reproducible on the same basis. Both DL4 families are generated from a seed, so an instance is named by its family and seed rather than stored; the accuracies in §16.2 and §16.6 are per-seed and can be recomputed instance by instance. The specification–key consistency tests of §16.4 are part of the suite rather than a one-off audit, and each was confirmed to fail when the defect it guards against was deliberately reintroduced — a test for a defect that has never been observed failing is an assertion of the same kind this section exists to distrust. The concordance audit of §16.3 requires no artefacts beyond two executors and the per-case responses, both of which the runner already records.

The three checks are released as runnable tools in the accompanying benchmark library rather than described only here, together with a written record of what they reported when first run against the library's previous version: five structural errors on a tree that its own validator passed with zero, of which four were bound instances byte-identical to an instance at a different level. A check that has only ever been run against the items it was written for has not been tested.

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A Compact Reporting Template

Core measured profile: $[C, I, O, T, H, SA]$ at reliability p
Conceptual profile: $[C, I, O, DI, SA]$, where $DI = F_A(H, p)$
Unified headline: $U_n\ [C_x, I_y, O_z, T_k/H_h, SA_s]$
Extended system profile: $[U;\ C, I, O, DI, SA;\ \lambda;\ D, A, R, V, S;\ \text{resources},\ \varphi]$

B Suggested Unified Promotion Tests

C Harness-Intelligence Reporting Template

- **Base LLM:** model id / version / hash.
- **Ladder tested:** identifier, and the generations included.
- **Per harness:** U^* | HLIS | $[C, I, O, T/H, SA]$ | reliability | bottleneck | false-completion rate | resource budget.
- **LLM summary:** benchmark vector $B(m)$ | Harness Scaling Curve | HIL-Level | HIL-Ceiling | Harness Gain | ρ_V | HIL-AUC and composite, if reported, with the cautions of §14.4 | HDS if harness-design tasks were run.
- **Application panel:** selected coordinates, reported separately from the common core.

Item-validity block (new in 1.6). Report beside every coordinate: the per-coordinate headroom (§13.15); the date of the last cross-tier concordance audit and the number of items it flagged; whether each parameterised item family carries a passing specification–key consistency test; and, for any item family whose score is graded rather than binary, the distribution of scores rather than the mean alone.

D Version History

Framework status. Level names and numerical thresholds remain proposed working definitions and should be revised when benchmark evidence warrants. Version 1.4 revised the score after running it; Version 1.6 revised the items after auditing them; Version 1.9 located headroom outside the mechanically-derivable envelope and revised the apparatus record after a retraction. Version 2.0 adds comparative evidence that the regime-switch family discriminates the two archived executor configurations on matched instances, while preserving the distinction between executor comparison and harness scaling.

Capability slice	Representative families	Primary HIL evidence	What the benchmark alone does not establish
Broad reasoning	HLE; MMLU-Pro	A_C breadth; hard closed-ended reasoning	Persistence, autonomy, organization, self-improvement, harness scaling.
Scientific reasoning	GPQA Diamond	A_C science	Longitudinal scientific learning or autonomous research execution.
Advanced mathematics	AIME-style; FrontierMath	A_C math	Project autonomy, tool orchestration, learning across episodes.
Coding	LiveCodeBench	A_C code plus bounded execution evidence	Repository-scale planning, persistence, intervention, organization.
Software engineering	SWE-bench and repo-level suites	DI at T2–T4; execution/verifier evidence	A general level unless intervention and cross-domain retention are controlled.
Abstract / adaptive	ARC-AGI-2/3	A_C fluid; DI exploration	Persistent individual or organizational evolution.
Tool calling	BFCL	DI tool-selection layer; abstention evidence	Long-horizon mission autonomy.
Interactive agent reliability	τ-bench family	DI success surface; reliability p	General cognition or self-improvement depth.
Research / browser agent	BrowseComp; GAIA	DI for research tasks; tool reach	Longitudinal learning, organizational evolution, self-model depth.
Vision + reasoning	MMMU; MathVista	A_C multimodal	Autonomous action, persistence, harness scaling.
Long context	RULER; LongBench	A_C context handling	**Not** I1: nothing in a single-episode retrieval score survives a process restart (§5.1).
Long-term memory	LoCoMo [15]; longitudinal dialogue and agent-memory suites	A_I evidence for M1–M2 when episodes are genuinely separated	Consolidation into procedure, and the governed self-improvement lineage M3 and above require.
Factuality	SimpleQA	A_C factuality/calibration	Autonomy, organization, self-improvement.
Real-user preference	Chatbot Arena	External preference evidence	A mechanistic or cumulative intelligence level.
Computer agent	OSWorld family	DI in executable GUI environments	Longitudinal I/O/SA maturity unless separately instrumented.

Executor	Rung	A_{DI} (v1.3)	$HLIS_{DI}$	False-done	Held back	Attempts
Family A, stronger	HG0	0.933	93.3	**2**	0	12
	HG1	0.933	93.3	0	2	12
	HG2	0.967	96.7	0	1	15
	HG3	0.967	96.7	0	1	14
Family B	HG0	0.933	93.3	**2**	0	12
	HG1	0.933	93.3	0	2	12
	HG2	0.967	96.7	0	1	16
	HG3	0.967	96.7	0	1	15
Family A, weaker	HG0	0.783	78.3	**4**	0	12
	HG1	0.917	91.7	0	3	12
	HG2	0.917	91.7	0	3	18
	HG3	0.933	93.3	0	2	14

Table 3: Three Harness Scaling Curves on ladder `dli-ladder/HG0-HG3`. 48 episodes per executor, all runs at H1. Families A and B are two separate vendor CLIs; absolute values are not a claim about any product.

	A, stronger		B		A, weaker	
Rung	v1.3	v1.4 net	v1.3	v1.4 net	v1.3	v1.4 net
HG0	93.3	**86.7**	93.3	**86.7**	78.3	**56.7**
HG1	93.3	93.3	93.3	93.3	91.7	91.7
HG2	96.7	96.7	96.7	96.7	91.7	91.7
HG3	96.7	96.7	96.7	96.7	93.3	93.3
Gain	3.3	**10.0**	3.3	**10.0**	15.0	**36.6**

	A stronger	B	A weaker	Reading
$HLIS_{DI}(HG0)$	93.3	93.3	78.3	the weaker executor starts far behind
HIL-Ceiling	**96.7**	**96.7**	93.3	and still ends behind
Harness Gain	3.3	3.3	**15.0**	but is **4.5× more harnessable**
HIL-AUC	**95.0**	**95.0**	88.8	AUC ranks by baseline, not by harness response

Executor	Gross HG0	Gross HG1	Difference	Net HG0 → HG1	False rejections
A, stronger	0.933	0.933	**0.000000**	0.867 → 0.933 (+0.067)	**0**
A, weaker	0.917	0.917	**0.000000**	0.833 → 0.917 (+0.083)	**0**

Table 4: Paired HG0/HG1 measurement. The gross figures are identical by construction; the entire contribution of the acceptance rung appears in the net figures.

Property	Typical benchmark score	HIL extension
Unit of analysis	Model or agent in one configuration.	Frozen pair $A(m,h)$, plus the same model across a ladder.
Autonomy	Implicit or benchmark-specific.	Explicit $T \times H \times p$ surface and frontier.
Human dependence	Not a first-class variable.	H0–H5 logged and constrained.
Cost of being wrong	Absent; pass or fail.	False completions priced at ρ (§13.9).
Persistence / learning	One episode, fixed state.	I-level tests require restart continuity and learning transfer.
Organization	Fixed agent architecture.	O-level tests evaluate coordination and reorganization with real role boundaries.
Self-awareness	Usually absent.	SA requires grounded state, limitation, causal-failure and lineage tests.
Model versus harness	Conflated in the final score.	HLIS measures a pair; Harness Gain and ρ_V separate model from harness unlock.

Layer	Report	Purpose
Contemporary benchmark layer	HLE/MMLU-Pro; GPQA; math; coding/SWE; ARC-AGI; BFCL; agent/research; multimodal; long context; factuality; preference; computer use — with exact versions.	Familiar capability slices; comparison with existing leaderboards.
HIL layer	A_C profile; $S_A(T,H)$ and $F_A(H,p)$; false-completion rate; I/O/SA levels; U^*; per-harness HLIS; the curve; HIL-Level; Ceiling; Gain; ρ_V; optional composite.	How the model becomes a persistent, delegated, organizational, self-modeling system as harness sophistication increases.

Coordinate	Why it remains outside core intelligence
Circle / λ	Measures which substrate improvement loops are reached or closed; substrate access is not cognition.
Write depth D	Measures which persistent cognitive states can be rewritten; permission to rewrite does not prove intelligent rewriting.
Authority A	Permission is not intelligence. A highly intelligent verifier may deliberately have almost no action authority.
Reach R	Access to files, tools, agents, robots, labs or networks increases possible action but is not cognition.
Verification V	Determines credibility of claims; it validates intelligence rather than constituting it.
Separation S	Organizational validity condition for proposer/evaluator/promoter roles.
Resources	Compute, memory, money, time and energy affect achievable performance but must not inflate the definition.
Physical floors φ	Irreducible latency and physics constraints rather than cognitive level.

Field	Example
Unified headline	U3
Core profile	C5, I4, O3, T4/H1, SA4
Reliability	$p = 0.80$
Delegation frontier	$F_A(H0, .80) = T3$; $F_A(H1, .80) = T4$
False-completion rate *(new in 1.4)*	0.00 at H1
Bottleneck	O3
Circle reach	$\lambda = (.42, .08, .03, 0, 0)$
Structural profile	D3, A2, V4, separated proposer/evaluator/promoter
Resource profile	declared compute, time, memory, energy/cost budget

Transition	Minimum new evidence in addition to full lower-level retention
U0 → U1	Persistent state and correct goal-level completion after restart or context discontinuity.
U1 → U2	Verified experience improves future performance on held-out related tasks; multi-step delegation under ≤H2.
U2 → U3	Governed self/organizational method improvement plus T3 strategy construction under ≤H1 and causal self-diagnosis.
U3 → U4	Validated improvement of the improvement process plus long-horizon T4 work and accurate modeling of self-change.
U4 → U5	Independent discovery of initially unknown solution method or knowledge, and self-directed experiments about world and self-unknowns.
U5 → U6	Mission → projects → tasks → execution over changing conditions with a persistent role/mission self-model.
U6 → UΩ	Repeated bounded-charter mission generation whose validated discoveries create new tools, agents or organizations that expand future reachable problems.

Version	Date	Principal change
1.0	Aug 2026	Five families, six coordinates, the gated U0–UΩ scale, HG0–HGΩ.
1.1	Aug 2026	HLIS for a pair; HIL for a base model across a ladder; the LLM scorecard.
1.2	Aug 2026	The Harness Scaling Curve; the contemporary-benchmark mapping.
1.3	Aug 2026	Formal HLIS: symbol definitions, achievement construction, the cumulative $q^* = \min$, weights, intervals, resource envelopes, and omit-don't-zero for N/A dimensions.
1.4	Aug 2026	The ladder built and a curve measured. A_{DI} redefined on delivered outcomes net of false completions (§13.9, Prop. 1); the run log must record acceptance (§13.12); ρ_V added (§14.3); two structural cautions on the composite (§14.4); the reference ladder recorded (§11.3); per-task specification status required; the self-certification rule for harness design stated explicitly; first measured curve and its limitations (§17.4).
1.5	Aug 2026	Long-term memory embedded inside Individual Intelligence as its temporal substrate, with a per-level M0–MΩ contract and long context excluded as evidence for I1; outcome completeness and the paired-rung control generalised into standing requirements.
1.6	Aug 2026	The items audited. Cross-tier concordance auditing required before failures are attributed to executors (§16.3); specification–key consistency tests required for parameterised item families (§16.4); graded item scoring with separated failure modes (§16.5); two DL4 item families built on specification scale and on search (§16.2); the finding that disclosed hazards do not desaturate a suite (§16.1); per-coordinate headroom and audit status added to the reporting standard (§13.15); the memory contract of 1.5 carried in (§5.1).
1.7	Aug 2026	GUI Perception added as a diagnostic subscale of C with a GP0–GP5 ladder and the $[P, G, S, A, R]$ computer-agent decomposition; visual-interface complexity added to task difficulty; the observation channel made part of the reported harness (§4.1).
1.8	Aug 2026	Memory Capability M separated from Individual Intelligence and measured independently as a supporting coordinate, joined to I by the one-way gate $I(A) \geq I_n \implies M(A) \geq \mu_n$; M excluded from the default HLIS product; M-Bench and the μ_n prerequisite table (§5.1).
1.9	Aug 2026	Four difficulty axes measured; the first one outside the envelope reaches the frontier; apparatus and item-validity findings carried forward in Version 2.0.
2.0	Aug 2026	Same-seed regime-switch comparison using the archived frontier and Haiku CSVs. Frontier passes 5/12 and Haiku 1/12; frontier RMSE is lower on 11/12. Exact two-sided McNemar $p = 0.125$ and sign-test $p = 0.00635$ are reported with conservative uncertainty and explicit limits. The comparison changes executor/model at fixed harness/protocol and is not an HSC segment.